A Model of Currency Linkages During Crises

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Abstract

This paper investigates why some global crises forge stable long-run linkages among major currencies while others fracture them. We address two important questions in currency markets during crises: (1) whether cointegration relationships of major currencies are preserved across distinct crisis episodes, and (2) whether long-horizon currency returns show lower correlation as currency autonomy declines. We present a model in which exchange rates load on global stochastic trends under autonomy. The model shows that the rank of the pass-through matrix together with an autonomy term determines whether cointegrating relations survive. We then take our model to market data and empirically analyze five major currency pairs, EUR, GBP, JPY, CHF, and AUD, against USD across the Dot-Com (1997-2000), Global Financial Crisis (2007–2009), and COVID-19 (2020–2022) crisis episodes. We find that the Global Financial Crisis uniquely produced a cointegrating relation, with deviations corrected by EUR/CHF/GBP over two- to five-week half-lives, while Dot-Com and COVID-19 displayed none. Our long-horizon correlations confirm that when weaker comovements emerge during a crisis, cointegration is weakened or vanishes. Our results also demonstrate that not all crises are born equal: only when shocks are both globally systemic and USD-centered, are stable long-run anchors preserved. Overall, we show that systemic USD-centered stress compresses autonomy, which causes currencies to load similarly and yields a stable anchor, while heterogeneous or novel persistent shocks dissolve this cointegration.

Keywords: currency linkages, foreign exchange risk, regime-dependent, financial crises, contagion.
JEL Classification: G15, G12, F31

1 Introduction

This paper investigates how global crises affect the long-run relationships among major currencies. Specifically, we address two questions: (1) Are cointegration relationships among currencies preserved during systemic crises? (2) How do long-horizon currency return correlations evolve as currency autonomy varies and cointegration evolves across market regimes?

Foreign exchange markets are the largest and most liquid over-the-counter financial markets, central to global trade, investment, and monetary transmission. Understanding the integration and comovement of currencies is vital for portfolio diversification, risk management, and tracing financial shock propagation. Episodes of systemic stress—most notably the Global Financial Crisis, the COVID-19 pandemic, and the Dot Com Boom—offer natural experiments to examine shifts in both short-run currency co-movements and their long-run equilibrium relationships.

Under frictionless, integrated markets, asset pricing theory predicts that bilateral exchange rates share a common global stochastic discount factor, implying cointegration with rank $N-1$ in an N-currency system Backus et al. (2001); Engel and West (2005). However, structural frictions, heterogeneous monetary regimes, and crisis conditions can alter these long-run linkages. Empirically, the Global Financial Crisis—characterized by USD funding stress—strengthened common global factor influences and cointegration, consistent with theory. In contrast, the COVID-19 period reveals a surprising breakdown of cointegration, suggesting a rise in currency-specific stochastic trends and autonomy that challenges conventional crisis integration narratives.

Our contribution is to unify factor-driven exchange rate models, no-arbitrage asset pricing, and Johansen cointegration analysis into a regime-dependent framework that allows for time-varying currency autonomy. We introduce an autonomy parameter $\psi_{i,r}$ governing the exposure of currency i to global factors in regime r. Crises like the Global Financial Crisis compress $\psi_{i,r}$ toward zero, reflecting diminished autonomy and tighter long-run linkages; by contrast, the COVID-19 crisis corresponds to increased autonomy and the emergence of new, independent stochastic trends, which fragment the cointegrating structure. This framework offers a structural explanation for the observed shifts in cointegration rank and currency comovement patterns.

Our stylized model links $\psi_{i,r}$ directly to the cointegration rank, adjustment speeds, and long-run correlation structures estimated from vector error correction models. Elevated global factor volatility and reduced autonomy yield higher long-run correlations and stronger network connectivity Flood and Rose (1995); Inoue and Rossi (2012), whereas increased autonomy leads to weaker cointegration and greater heterogeneity across currencies—empirically confirmed during COVID-19.

Our work bridges structural asset-pricing theory with empirical cointegration analysis, applying this synthesis to both major and emerging foreign exchange markets across multiple crises. While prior research emphasizes short-run volatility spillovers Melvin and Taylor (2009); Aloui et al. (2014) or regional currency blocs Kanas (2005), we provide a comprehensive long-run equilibrium perspective with regime-dependent factor loadings and network diagnostics. Moreover, we document and structurally interpret the COVID-19-induced breakdown of long-run parity—a novel phenomenon absent in prior crises and largely unexamined in cointegration literature.

Our framework distinguishes between the existence of long-run equilibrium relationships (cointegration) and the degree of long-horizon return comovement. While cointegration imposes a structural condition on the number of independent stochastic trends, long-horizon return correlation reflects how tightly currencies co-move over extended horizons—even in the absence of cointegration. This distinction allows us to show that COVID-19 weakened structural linkages (cointegration), yet certain currencies continued to exhibit substantial long-horizon comovement, pointing to a regime where currencies load differently—though not independently—on evolving global risk factors.

Our analysis uncovers a novel structural phenomenon in the behavior of major foreign exchange markets during global crises. While prior research largely documents increased integration or temporary disruptions in cointegration during episodes such as the Global Financial Crisis, we find that the COVID-19 pandemic introduced a new global stochastic trend factor—distinct and independent from pre-existing common risk factors. This emergence increased the number of independent stochastic trends in the system, thereby eroding the long-run cointegration relationships among major currencies.

Crucially, we show that the presence of an additional stochastic trend does not necessarily weaken or destroy cointegration if the new trend is spanned by existing global factors. However, the COVID-19 crisis was characterized by a truly novel factor that could not be represented as a linear combination of prior global factors. This structural break led to the observed breakdown of long-run linkages, signaling a regime shift in the underlying global risk environment.

This finding challenges the prevailing crisis-contagion narrative, which assumes crises uniformly compress currency autonomy and strengthen long-run ties. Instead, our results highlight that crises can reshape the fundamental factor structure of currency markets in qualitatively different ways, with significant implications for asset pricing, risk management, and international portfolio diversification.

Previous studies on foreign exchange market integration during crises have largely focused on shifts in short-run volatility spillovers Melvin and Taylor (2009); Aloui et al. (2014) or the temporary breakdown and subsequent restoration of long-run parity Flood and Rose (1995); Dungey et al. (2005). While these works document changes in cointegration rank or correlation patterns, they do not explicitly characterize the factor structure changes underpinning these shifts.

Our contribution lies in combining a structural asset pricing framework with cointegration analysis to demonstrate that a crisis can introduce genuinely new global risk factors, thereby increasing the number of independent stochastic trends. This insight deepens the understanding of how crisis regimes can lead to permanent structural breaks rather than transient deviations in exchange rate dynamics.

To the best of our knowledge, this is the first paper to document and interpret the breakdown of long-run parity among major currencies during COVID-19 as the result of an emergent, distinct global stochastic trend and/or a rise in autonomy such that all currencies do not react strongly in unision in their responses to global common factors. This enriches the literature by providing a mechanistic explanation for the observed heterogeneity in foreign exchange market responses during different global crises.

1.1 The relevance of the foreign exchange market

The foreign exchange market is uniquely suited to study long-run cointegration and crisis dynamics for several reasons. First, foreign exchange markets are the largest and most liquid financial markets globally, with daily turnover exceeding that of equity and bond markets combined, ensuring that prices reflect real-time global information with minimal frictions. Second, unlike equities and bonds, foreign exchange rates are inherently cross-border relative prices directly tied to macroeconomic fundamentals, monetary policy, and risk premia across countries. This relative pricing structure creates a natural environment to test long-run equilibrium relationships and cointegration, grounded in no-arbitrage and uncovered interest parity conditions, which are less cleanly applicable to segmented or jurisdiction-bound markets such as equities or sovereign bonds. Third, foreign exchange markets operate continuously across global time zones, enabling comprehensive high-frequency data coverage necessary to detect subtle regime shifts and structural changes, especially during crises. Finally, the dominant role of the U.S. dollar as a global funding currency amplifies the transmission of global risk shocks, making foreign exchange markets a natural laboratory to investigate the interplay between global systemic factors and currency-specific autonomy during extreme events. Together, these features establish foreign exchange markets as an ideal setting to study regime-dependent cointegration and crisis-induced market dynamics, offering insights that are less readily obtained from equity or bond markets.

In summary, this paper addresses two important questions in foreign exchange markets during crises: (1) whether cointegration relationships among major currencies persist across distinct crisis episodes, and (2) whether long-horizon currency returns show lower or higher correlation as currency autonomy declines (increases) and evidence of cointegration strengthens (weakens). We analyze daily spot rates for five major currency pairs (EUR/USD, USD/JPY, GBP/USD, USD/CHF, AUD/USD), focusing on the the Dot Com Boom (1997-2000), the Global Financial Crisis (2007–2009), and the COVID-19 pandemic (2020–2022). Our findings show no evidence of long-run cointegration during the Dot-Com crisis, indicating that currencies saw increased autonomy amid volatility. We confirm the existence of cointegration among major currencies during the Global Financial Crisis, consistent with reduced currency autonomy driven by common global risk factors. In contrast, the COVID-19 period exhibits a breakdown of cointegration amid elevated volatility, driven by the emergence of a new, distinct global stochastic trend which increases the number of independent stochastic trends and erodes long-run linkages, reflecting greater currency-specific autonomy. These results challenge the standard crisis-contagion narrative by demonstrating that global crises may not uniformly increase foreign exchange integration. We highlight the importance of regime-dependent pass-through parameters in exchange rate models, discuss the relevant implications and explain the condition under which cointegration can be preserved across crisis periods.

2 Literature review

2.1 Long-run comovement and cointegration in foreign exchange

A foundational strand of the foreign exchange literature documents common stochastic trends among major currency pairs, employing cointegration methods to study long-run parity relations. Early work establishes that several dollar exchange rates share persistent common drivers, motivating vector error correction models and rank tests to identify cointegration relationships Baillie and Bollerslev (1989); Johansen (1988, 1991). Complementing this, asset-pricing perspectives reconcile the apparent near-random-walk behavior of exchange rates at short horizons with the relevance of fundamentals in present-value terms Engel and West (2005). These insights underpin the widespread use of Johansen's framework to test whether N dollar exchange rates admit $N-1$ cointegrating relations in integrated FX markets. However, most of these studies predate the COVID-19 crisis and do not address how new global factors can alter the number of stochastic trends, which is central to our analysis.

2.2 Risk-based and intermediary-based views of foreign exchange

Modern asset pricing models link exchange rates to priced risk factors and intermediary constraints. Currency returns load on common factors such as the "dollar" and "carry," which help explain the cross-section of currency risk premia Lustig et al. (2011). Meanwhile, intermediary asset pricing models emphasize that exchange rates also respond to dealers' balance-sheet constraints and shifts in international liquidity; these frictions can alter currency comovement by repricing currencies and affecting funding conditions Gabaix and Maggiori (2015). These approaches provide structural channels—risk prices and funding constraints—through which global shocks synchronize currencies while allowing heterogeneity across countries and over time.

2.3 Crises, funding stress, and the global dollar system

Crisis episodes reveal how dollar funding conditions transmit globally through foreign exchange markets. The Global Financial Crisis impaired USD funding markets, causing sharp deviations in foreign exchange swap pricing, heightened margins, and a shortage of dollars outside U.S. banks Baba and Packer (2009); McGuire and von Peter (2009). These dislocations manifested as violations of covered interest parity, which subsequent research modeled as equilibrium phenomena linked to dealer leverage and regulatory constraints Du et al. (2018). These mechanisms imply that under stress, currencies may load more strongly and more similarly on the global dollar factor, potentially compressing long-run autonomy even amid turbulent short-run dynamics.

2.4 COVID-19 as a distinct stress event

The COVID-19 pandemic triggered an extraordinary "dash for dollars," with severe stress in foreign exchange swap markets and unprecedented policy interventions including central bank swap lines and liquidity facilities. Bank for International Settlements analyses document acute dislocations in March 2020 followed by normalization as swap lines expanded Schrimpf et al. (2020). Related work shows that swap lines mitigated dollar funding shortages and narrowed covered interest parity deviations across jurisdictions Bahaj and Reis (2020). While these studies emphasize short-run funding stresses, they raise important questions about whether COVID also altered *long-run* currency linkages in ways distinct from the Global Financial Crisis.

2.5 Structural breaks, regime dependence, and inference

From a time-series perspective, identifying regime shifts in cointegration relationships is critical. Residual-based tests permit endogenous single breaks in cointegrating vectors Gregory and Hansen (1996), while multiple-break frameworks support inference in high-dimensional systems Qu and Perron (2007). These tools complement Johansen's cointegration tests to assess whether cointegration weakens, vanishes, or re-emerges across crisis regimes such as the Global Financial Crisis and COVID.

2.6 Connectedness and network approaches

A complementary literature employs variance-decomposition-based connectedness metrics Diebold and Yilmaz (2012) and network-theoretic methods (e.g., minimum spanning trees, Planar Maximally Filtered Graphs) to measure the strength and direction of cross-market spillovers. These approaches distinguish the *number* of common stochastic trends from the *tightness* of co-movement (covariance or connectedness), offering nuanced insights into crisis dynamics.

2.7 How this paper advances the literature

This paper contributes along four key dimensions:

2.7.1 Unified rheory and empirics across regimes

We embed a no-arbitrage, log-normal stochastic discount factor with nonstationary global factors into a pass-through framework that directly maps to Johansen cointegration and vector error correction model objects. This framework nests the classic $N-1$ cointegration benchmark and also accommodates multi-trend regimes where rank may decline or increase, capturing scenarios often missed by correlation or short-run spillover-based analyses (cf. Du et al. (2018); Johansen (1991)).

2.7.2 Structural interpretation of autonomy

We introduce currency and regime specific autonomy parameters governing factor pass-through, providing a structural interpretation of cross-country heterogeneity highlighted in emerging markets resilience studies Aizenman and Jinjarak (2014) and observed comovement dynamics during both the Global Financial Crisis and COVID Baba and Packer (2009); Schrimpf et al. (2020).

2.7.3 Distinguishing rank from tightness

We separate the *number* of common stochastic trends (cointegration rank) from the *tightness* of long-run co-movement (long-run covariance and connectedness). This distinction clarifies how a crisis can simultaneously exhibit high correlation yet *lower* cointegration rank if additional nonstationary global factors emerge—an insight supported by intermediary-based models and funding market evidence.

2.7.4 COVID as a structural outlier

While the Global Financial Crisis reflects compressed autonomy and strengthened long-run cointegration, we document that the COVID-19 crisis coincided with weaker cointegration among major currencies. This is consistent with the emergence of new global stochastic trends, distinct funding mechanisms, and novel policy backstops (e.g., swap lines), thereby updating the crisis-integration paradigm in light of modern dollar funding architecture Schrimpf et al. (2020).

Together, these contributions bridge classic cointegration literature Baillie and Bollerslev (1989); Johansen (1988, 1991) with contemporary finance theory—incorporating risk factors, intermediary frictions, and policy interventions—to provide a structural account of why long-run foreign exchange linkages can strengthen during one crisis yet weaken in another.

3 Model Setup

We now present a regime-dependent factor model linking the long-run dynamics of N non-USD exchange rates to a set of nonstationary global factors. The U.S. is the numeraire. Let the spot exchange rate be

$$S_{i,t} = \text{units of currency } i \text{ per USD} \quad (i = 1, \dots, N),$$

so an increase in $S_{i,t}$ indicates USD appreciation against currency i. Define $X_{i,t} = \log S_{i,t}, \quad X_t = (X_{1,t}, \dots, X_{N,t})'$.

3.1 Pricing kernel stochastic discount factor

We price assets in USD using a log-normal stochastic discount factor:

$$\log M_{t+1} = -r_t - \tfrac{1}{2}\,\varsigma_r^2 - \xi_{t+1}, \tag{1}$$

where r_t is the USD short rate, ς_r^2 is the total SDF innovation variance, ξ_{t+1} is the stochastic discount factor innovation, and $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$ by construction.

3.1.1 Multi-factor global structure

In regime r, the innovation admits q_r nonstationary global factors:

$$\xi_{t+1} = \lambda_r^\top \Delta g_{t+1} + \upsilon_{t+1}, \quad \mathbb{E}_t[\upsilon_{t+1}\Delta g_{t+1}] = 0. \tag{2}$$

where:

- $g_t = (g_{1,t}, \ldots, g_{q_r,t})'$ are q_r independent integrated of order one global factors, e.g., USD funding stress, global risk appetite, commodity-price shocks.
- $\lambda_r \in \mathbb{R}^{q_r}$ are stochastic discount factor level factor loadings in regime r.
- $\upsilon_{t+1} \sim \mathcal{N}(0, \sigma_\upsilon^2)$ is an idiosyncratic USD pricing-kernel shock orthogonal to g_{t+1}.

Each factor follows a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_k,r}\,\varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \quad k = 1, \ldots, q_r. \tag{3}$$

The total stochastic discount factor innovation variance is:

$$\varsigma_r^2 = \lambda_r'\,\Sigma_{g,r}\,\lambda_r + \sigma_\upsilon^2, \quad \Sigma_{g,r} = \mathrm{diag}(\sigma_{g_1,r}^2, \ldots, \sigma_{g_{q_r},r}^2).$$

The currencies each have varying exposures to the vector of global factors (e.g., global risk, liquidity conditions, macroeconomic policy shifts), which means that the common-factor block loads differently on each currency: for some currencies, one or more factor loadings are large (high coefficients along specific factor directions), while for others they are small or negligible. In (2), the innovation to the stochastic discount factor is decomposed into two conceptually distinct components: a *systematic* term, $\lambda_r^\top \mathbf{g}_{t+1}$, and an *idiosyncratic* term, υ_{t+1}. The q_r-dimensional vector $\mathbf{g}_{t+1}$ captures shocks that are common—though not necessarily equally important—to all currencies; examples include shifts in USD funding conditions, coordinated changes in major central bank policy stances, or global risk-on/risk-off episodes. The vector λ_r measures the sensitivity of the *pricing kernel* to each of these common shocks in regime r, and in the affine solution each currency i inherits its own q_r-vector of pass-through coefficients $\lambda_{i,r}$, reflecting its specific exposure profile to the different global factors. By contrast, υ_{t+1} represents purely currency-specific noise in the stochastic discount factor, orthogonal to $\mathbf{g}_{t+1}$ by construction. Its variance σ_υ^2 fully determines its scale, so no additional factor loadings are needed at the stochastic discount factor level. Economically, the vector λ_r captures how each orthogonal component of the common-factor block propagates through the pricing kernel, while the heterogeneous $\lambda_{i,r}$ map those common shocks into individual currency returns; the idiosyncratic term υ_{t+1} is already asset-specific and its effects are entirely absorbed by the currency it directly influences.

3.2 Payoffs, no-arbitrage and an affine solution

A one-period claim pays $D_{i,t+1}$ units of currency i; its USD price is

$$P_{i,t} = \mathbb{E}_t \left[M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right]. \tag{4}$$

For a pure foreign exchange claim delivering one unit of currency i (i.e. $D_{i,t+1} = 1$), replication implies $P_{i,t} = 1/S_{i,t}$. No-arbitrage gives:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

or equivalently,

$$1 = \mathbb{E}_t \left[M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right] = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right], \tag{5}$$

our key foreign exchange pricing restriction. The intuition behind the no riskless arbitrage condition is that the dollar price paid today for a future claim of one unit of foreign currency

$$\mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

must equal the current dollar cost of acquiring that same unit of foreign currency:

$$\frac{1}{S_{i,t}}.$$

If this equality failed, arbitrageurs could construct riskless profit opportunities by trading currencies and bonds, contradicting the absence of arbitrage in equilibrium. Risk-bearing profits remain possible but are not part of this no-arbitrage condition.

3.3 Solving for exchange-rate levels

Assuming (i) the stochastic discount factor is logarithmically normal as in (1)-(2) and (ii) the fundamentals z_t ($K \times 1$) are stationary and independent of global factor innovations, the logarithmic linear solution takes the affine form

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r}\, g_t + \Gamma'_i z_t + u_{i,t}, \tag{6}$$

with $u_{i,t}$ stationary, and

$$\mu_{i,r} := -\bar{r}_r - \tfrac{1}{2}\varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \kappa_{i,r}, \tag{7}$$

$$\kappa_{i,r} \equiv \tfrac{1}{2}\big[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\upsilon^2 + \mathrm{Var}_t\big(\Delta(\Gamma_i^\top z_t + u_{i,t})\big)\big]\,, \tag{8}$$

$$\Gamma'_i z_t = \sum_{k=1}^{K} \gamma_{i,k}\, z_{k,t}, \quad z_t = (z_{1,t}, \ldots, z_{K,t})' \text{ is stationary,} \tag{9}$$

$$u_{i,t} = X_{i,t} - \mu_{i,r} - \lambda'_{i,r} g_t - \Gamma'_i z_t, \quad u_{i,t} \text{ stationary.} \tag{10}$$

Here $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$ is the regime-r mean short rate; $(\mu_r, \Sigma_{g,r}, \lambda_r)$ are regime parameters of the q_r-dimensional global-factor process and stochastic discount factor; and σ_υ^2 is the variance of the stochastic discount factor idiosyncratic residual. The global-factor vector g_t enters linearly in the pricing kernel, so $X_{i,t}$ must load linearly on g_t. Currencies differ in their exposure profiles to the q_r orthogonal global factors, with some exhibiting large loadings in certain factor directions and others having smaller or near-zero loadings. This heterogeneity is captured by the currency-specific vector $\lambda_{i,r}$, while λ_r measures the sensitivity of the pricing kernel to each global factor in regime r. In the stochastic discount factor, the innovation decomposition

$$\xi_{t+1} = \lambda_r^\top \Delta g_{t+1} + \upsilon_{t+1}$$

separates the *systematic* component, $\lambda_r^\top \Delta g_{t+1}$, from the *idiosyncratic* component, υ_{t+1}. The vector Δg_{t+1} represents shocks common to all currencies—such as shifts in global risk appetite, changes in major reserve currency funding conditions, or synchronized monetary policy moves—but whose economic impact varies across currencies according to their $\lambda_{i,r}$ profiles. By contrast, υ_{t+1} is orthogonal to Δg_{t+1} by construction, representing purely currency-specific noise in the stochastic discount factor; its variance σ_v^2 fully determines its scale, so no additional factor loadings are needed for this term at the SDF level.

We note that the use of Δg_{t+1}, rather than g_{t+1}, ensures that SDF innovations are stationary. Since g_t is assumed to be integrated of order one with stationary increments, modeling shocks in terms of Δg_{t+1} prevents the SDF from inheriting nonstationarity and preserves internal consistency with the cointegration structure.

The vector z_t is taken to be stationary so that exchange rates can load on *mean-reverting* conditions without introducing additional stochastic trends. While short-horizon exchange rates often approximate random walks—weakening their level relations with trending fundamentals[1]—stationary variables capture time-varying but transitory forces that help explain cross-sectional heterogeneity and conditional dynamics.[2] Treating z_t as stationary preserves the common integrated of order one trends driven solely by g_t and leaves the cointegration rank determined by the rank of Λ_r; standard vector error correction models or cointegrated vector autoregression frameworks allow such stationary exogenous controls without affecting rank tests.[3]

Finally, the residual $u_{i,t}$ measures the deviation of $X_{i,t}$ from the level implied by its long-run relation with g_t and z_t. If $u_{i,t}$ were nonstationary, it would imply an additional persistent factor orthogonal to g_t, contradicting the assumption that g_t spans all nonstationary variation. Stationarity of $u_{i,t}$ ensures that any cointegration between $X_{i,t}$ and g_t is not impaired, and confines $u_{i,t}$ to mean-reverting, thereby preserving the internal consistency of the model's long-run factor structure. The USD price of one unit of currency i delivered at $t+1$ is $1/S_{i,t}$. No-arbitrage implies:

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

Solving under log-normality and stationary fundamentals z_t yields:

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}' \, g_t + \Gamma_i' z_t + u_{i,t}, \tag{11}$$

where $\lambda_{i,r} \in \mathbb{R}^{q_r}$ reflects currency-specific exposure to the q_r global factors, $g_t \in \mathbb{R}^{q_r}$. We now present this result more formally in the lemma that follows below.

[1] See Engel and West (2005) for the near–random-walk implication; classic out-of-sample evidence includes Meese and Rogoff (1983).

[2] Taylor-rule fundamentals and related stationary deviations have documented predictive content at certain horizons; see, e.g., Molodtsova and Papell (2009).

[3] See Johansen's vector error correction model framework as commonly implemented.

Lemma 1. *(Affine Representation of Exchange Rates) Suppose the log exchange rate* $X_{i,t} = \log S_{i,t}$ *satisfies the difference equation*

$$\Delta X_{i,t+1} = a_{i,r} + \lambda_{i,r}'\,\Delta g_{t+1} + \Gamma_i'\,\Delta z_{t+1} + \epsilon_{i,t+1}, \tag{12}$$

together with the no-arbitrage restriction

$$\mathbb{E}_t\left[M_{t+1}e^{X_{i,t}-X_{i,t+1}}\right] = 1.$$

Assume the following:

A1 *Log-normal Stochastic Discount Factor: The stochastic discount factor satisfies*

$$\log M_{t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad \xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2),$$

with $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

A2 *Stochastic Discount Factor Shocks: The innovation is driven by global factor shocks,*

$$\xi_{t+1} = \lambda_r'\Delta g_{t+1} + \upsilon_{t+1},$$

with υ_{t+1} *stationary and orthogonal to* Δg_{t+1}.

A3 *Integration Orders:* g_t *is integrated of order one with stationary increments* Δg_{t+1}*, while* z_t *is integrated of order zero.*

A4 *Orthogonality:* $\{\Delta g_{t+1}, \Delta z_{t+1}, \epsilon_{i,t+1}, \upsilon_{t+1}\}$ *are mean-zero, jointly Gaussian, with*

$$\mathbb{E}_t[\Delta z_{t+1}\,\Delta g_{t+1}'] = 0, \quad \mathbb{E}_t[\epsilon_{i,t+1}\,\Delta g_{t+1}'] = 0, \quad \mathbb{E}_t[\epsilon_{i,t+1}\,\Delta z_{t+1}'] = 0.$$

Then the log exchange rate admits the affine representation

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}'\,g_t + \Gamma_i' z_t + u_{i,t},$$

where $u_{i,t}$ *is integrated of order zero and* $\mu_{i,r}$ *absorbs constants including the risk-adjusted drift identified from the no-arbitrage restriction. Note, a formal proof is provided in the Appendix.*

3.4 Regime-dependent pass-through ("autonomy")

In order to model how the sensitivity of exchange rates to the global factor vector changes across regimes, we parameterize the factor loadings for currency i in regime r as

$$\lambda_{i,r} = \big(1 - \psi_{i,r}\big) \odot \bar{\lambda}_i, \qquad \bar{\lambda}_i \neq 0 \text{ fixed}, \tag{13}$$

where $\psi_{i,r} = (\psi_{i1,r}, \ldots, \psi_{iq_r,r})' \in [0,1]^{q_r}$ is the vector of "autonomy" parameters of currency i from each global factor in regime r, $\bar{\lambda}_i \in \mathbb{R}^{q_r}$ is the baseline (regime-invariant) vector of factor loadings, and $\odot$ denotes the Hadamard (elementwise) product. A lower $\psi_{ik,r}$ implies stronger pass-through from factor k (larger $|\lambda_{ik,r}|$), meaning that the currency's value is more sensitive to movements in that global factor. A higher $\psi_{ik,r}$ implies greater autonomy from factor k in that regime. The following assumptions describe how volatility and autonomy evolve across regimes, and ensure tractability of the model:

B1 Crisis volatilities: $\sigma^2_{g_k,\text{crisis}} \geq \sigma^2_{g_k,\text{no}-\text{crisis}}$ for all k, with strict inequality for at least one k. *Crisis periods exhibit heightened volatility in at least one global factor, reflecting increased uncertainty and tighter global financial conditions.*

B2 Autonomy compression: $\psi_{ik,\text{crisis}} \leq \psi_{ik,\text{no}-\text{crisis}}$ for all i, k. *Market stress and funding pressures force more synchronized exchange-rate responses to the global factors.*

B3 Stationarity and orthogonality: z_t and u_t are stationary and orthogonal to factor innovations ε_{t+1}. *This preserves the cointegration rank implied by Λ_r and maintains orthogonality between predictable components and shocks.*

3.4.1 Factor-loading structure

For each regime r, define the $N \times q_r$ pass-through matrix

$$\Lambda_r \;=\; \begin{bmatrix} \lambda'_{1,r} \\ \vdots \\ \lambda'_{N,r} \end{bmatrix} \;=\; \begin{bmatrix} (1-\psi_{11,r})\,\bar{\lambda}_{11} & \ldots & (1-\psi_{1q_r,r})\,\bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1-\psi_{N1,r})\,\bar{\lambda}_{N1} & \ldots & (1-\psi_{Nq_r,r})\,\bar{\lambda}_{Nq_r} \end{bmatrix},$$

which stacks the currency-specific loading vectors $\lambda'_{i,r}$ across i.

Here:

- $\lambda_{i,r} \in \mathbb{R}^{q_r}$ are the currency-specific loadings on the q_r global factors in regime r.

- $\psi_{ik,r}$ is the "autonomy" of currency i from factor k in regime r; lower values imply higher pass-through from that factor.
- Γ_i loads on stationary fundamentals z_t, which do not alter the cointegration rank.
- $u_{i,t}$ is stationary and mean-reverting.

3.4.2 Exchange-rate system

With these definitions, the cross-section of exchange rates in regime r is

$$X_t = \mu_r + \Lambda_r \, g_t + \Gamma \, z_t + u_t, \tag{14}$$

where $\mu_r \in \mathbb{R}^N$ collects the regime-specific intercepts, $g_t \in \mathbb{R}^{q_r}$ is the vector of nonstationary global factors, Γ is the $N \times K$ matrix of stationary-factor loadings, and u_t is the vector of stationary residuals.

3.5 Cointegration rank and regime dependence

Given the affine representation

$$X_t \;=\; \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

the long-run dimension of comovement among the N exchange rates in regime r is determined by the rank of the pass-through matrix Λ_r. The vector $g_t \in \mathbb{R}^{q_r}$ collects the q_r nonstationary global factors relevant in regime r, while $\Lambda_r \in \mathbb{R}^{N \times q_r}$ stacks the currency-specific loading vectors $\lambda'_{i,r}$. Stationary fundamentals z_t and residuals u_t do not affect the cointegration rank because they are integrated of order zero. The number of distinct common stochastic trends transmitted to X_t in regime r is

$$\operatorname{rank}(\Lambda_r) \;\leq\; \min(N, q_r).$$

This yields the standard cointegration rank result:

- If $\operatorname{rank}(\Lambda_r) = 1$, there is a *single* nonstationary common component, and the system has $N - 1$ cointegrating relations (the single-factor baseline).
- If $\operatorname{rank}(\Lambda_r) = k > 1$, there are $N - k$ cointegrating relations (multiple independent global trends drive the system).

- If $\text{rank}(\Lambda_r) = N$, there is no cointegration (all stochastic trends are common and equal to the number of currency variables, hence it is not possible to write the currencies in linear combinations in a way that eliminate all the common stochastic trends. Hence no cointegration).

- If $\text{rank}(\Lambda_r) = 0$, there is no common stochastic trend (either all trends are indiosyncratic, so each currency contains its own idiosyncratic stochastic trend) and not feasible to eliminate the stochastic trends so that the currencies are written as linear combinations. Hence no cointegration. Or there are no stochastic trends at all (if there are no idiosyncratic trends), in which case the currency variables are stationary, with no integer integration, and hence cannot be cointegrated.

3.5.1 Regime variation

Both the factor volatilities $\sigma^2_{g_k,r}$ and the autonomy parameters $\psi_{ik,r}$ can vary across regimes, altering the effective rank of Λ_r. For example:

- A crisis may *increase* similarity in response to global factors (making common trends more dominant) and/or *compress* autonomy parameters (aligning currencies more closely with the factors).

- If a new factor emerges—e.g., a pandemic-specific risk driver during COVID—it can reduce cointegration relations by raising autonomy. In the extreme, if enough independent trends are introduced, cointegration may disappear even if it was present.

3.6 Main propositions

The following propositions summarize the key theoretical implications of the multi-factor, regime-dependent pass-through model developed above. They connect the statistical properties of the pass-through matrix Λ_r to observable patterns of cointegration, long-run comovement, and regime shifts in exchange-rate dynamics. Formal proofs are provided in the Appendix.

3.6.1 Rank–cointegration link

Proposition 1. *Given regime r and assumptions B1–B3, the cointegration rank of X_t is*

$$\textit{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

Economic interpretation: This result formalizes the direct link between the number of independent nonstationary global trends and the existence of long-run exchange-rate linkages. If $\text{rank}(\Lambda_r) = 1$, then all persistent variation in each exchange rate is driven by a single common global factor (e.g., the USD funding/liquidity factor), yielding $N-1$ cointegrating relationships, meaning the exchange rates have established long run relationships and are related in the long run in $N-1$ different ways, where in each case the residual is stationary, with the residual being the spreads between the exchange rate normalized on and a linear combination of the other exchange rates. As $\text{rank}(\Lambda_r)$ increases, additional nonstationary drivers (for example such as stochastic factors like commodity-price trends, regional funding shocks, or segmented risk premia) introduce new sources of long-run divergence, reducing the number of cointegrating relations. In the extreme case $\text{rank}(\Lambda_r) = N$, each currency has its own independent stochastic trend, they do not share any common stochastic trend, and hence no long-run relationship survives as no combination of something that do not share anything in common would yield any established stable long run relationship.

If $\text{rank}(\Lambda_r)$ remains finite and bounded during crisis, such that (i) $\sigma^2_{g_k,r}$ rises for one or more factors, and (ii) $\lambda_{ik,r}$ is non-negligible and similar across currency for each global factor, i.e. react similarly to each global factor and the response is non-trivial, which means there is non-zero convergence in response to the global factors across currencies (autonomy compression, each currency has less autonomy from global factors, that is, the common stochastic trends), then long-run correlations, that is the correlation of long-run currency returns across currencies, increases, and factor volatility explains a larger share of the total variance of the long horizon returns for each currency.

3.6.2 Currency level cointegration and co-movement of long horizon currency returns

Proposition 2. *For currencies $i = 1, \ldots, N$ and common stochastic trends $k = 1, \ldots, q_r,\ q_r \leq N$,*

$$\begin{aligned} &\lambda_{1k} \approx \lambda_{2k} \approx \cdots \approx \lambda_{Nk}, \\ \forall\, i \neq j,\ \forall\, k \in \{1, \ldots, q_r\} : \quad &\lambda_{ik} \approx \lambda_{jk}, \quad \lambda_{ik} \neq 0,\ \lambda_{jk} \neq 0. \end{aligned}$$

then the correlation of long-horizon returns between currencies i and j increases toward one:

$$\rho_{ij}(H) \equiv \text{Corr}\big(r_{i,H}, r_{j,H}\big) \ \to\ 1,$$

where $r_{i,H}$ denotes the H-period return of currency i.

Remark: When the long-horizon correlation among currency returns is low and moves away from one, this suggests heterogeneous responses to the common stochastic trends (currencies have greater autonomy from the common stochastic trends). Increased autonomy weakens the shared component and, in practice, can diminish—or even eliminate—the likelihood of cointegration across the set. Low correlation does not preclude cointegration, but it weakens the case for cointegration, i.e. a tight long-run equilibrium. This is tested for empirical evidence in the empirical section[4]

Economic interpretation: Crises amplify co-movement through two reinforcing channels. First, the variance channel operates when higher volatility in the common factors increases their share of total long-horizon variance. Second, the loading-alignment channel arises when lower dispersion in $\psi_{ik,r}$ makes currency responses more similar in both magnitude and sign, reducing idiosyncratic offsets. Even without introducing new stochastic trends, these channels generate tighter long-run alignment—a pattern consistent with the "risk-on/risk-off" behavior observed during global financial stress.

3.6.3 Rank Change in Crisis

Proposition 3. *If new* ***persistent global shocks*** *emerge, or if* ***autonomy shifts*** *driven by, for example, the emergence of a never-before seen idiosyncratic stochastic trend, such that*

$$\operatorname{rank}(\Lambda_{crisis}) > \operatorname{rank}(\Lambda_{non\text{-}crisis}),$$

then the number of cointegrating relationships falls. In the extreme case where $\operatorname{rank}(\Lambda_{crisis}) = N$*, cointegration disappears entirely.*

Economic interpretation: Certain crises—such as the COVID-19 shock—introduce structural changes that are *not spanned* by existing common stochastic trend global factors, such as pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks. If these shocks are persistent, load heterogeneously across currencies, and are linearly independent of existing global factors, then they increase the effective dimension of the global factor process g_t, either raising the rank of Λ_r by introducing a new common stochastic trend, or/and increasing the degree of autonomy (for example, by introducing an idiosyncratic stochastic trend) so that all currencies don't all load the same. This weakens cointegration, and in the limiting case of full rank of common stochastic trend ($\operatorname{rank}(\Lambda_r) = N$), a complete breakdown of cointegration across currencies.

[4] For formal inference, cointegration tests remain decisive; the correlation-based argument here is interpretive.

4 Data

Our dataset consists of daily spot exchange rates, covering recognized crisis periods in the global market. All exchange rates are quoted against the U.S. dollar. Prices are transformed to natural logarithms for cointegrated analysis, and returns are calculated as first differences of log prices for the analysis of the correlation of long run currency returns. We focus on the most liquid "major" currency pairs, representing a blend of global reserve currencies, safe-haven assets, and commodity-linked units:

- **EUR/USD** — Euro, the second most important reserve currency and the deepest non-USD FX market.
- **GBP/USD** — British pound, historically significant but subject to geopolitical regime shifts (e.g., Brexit).
- **USD/JPY** — Japanese yen, a major safe-haven asset with persistent interest rate differentials.
- **USD/CHF** — Swiss franc, another canonical safe-haven currency.
- **AUD/USD** — Australian dollar, a commodity-linked currency influenced by terms-of-trade shocks.

4.0.1 Crisis Subperiods

To examine stability and regime dependence in long-run currency relationships, we split the sample into three key subperiods:

- **Dot-Com Crisis:** May 1997 – June 2000.
- **Global Financial Crisis (GFC):** August 2007 – June 2009.
- **COVID-19 Crisis:** Core pandemic phase from February 2020 – December 2021,

These subperiods align with macro-financial chronology and allow direct comparison of cointegration stability across regimes.

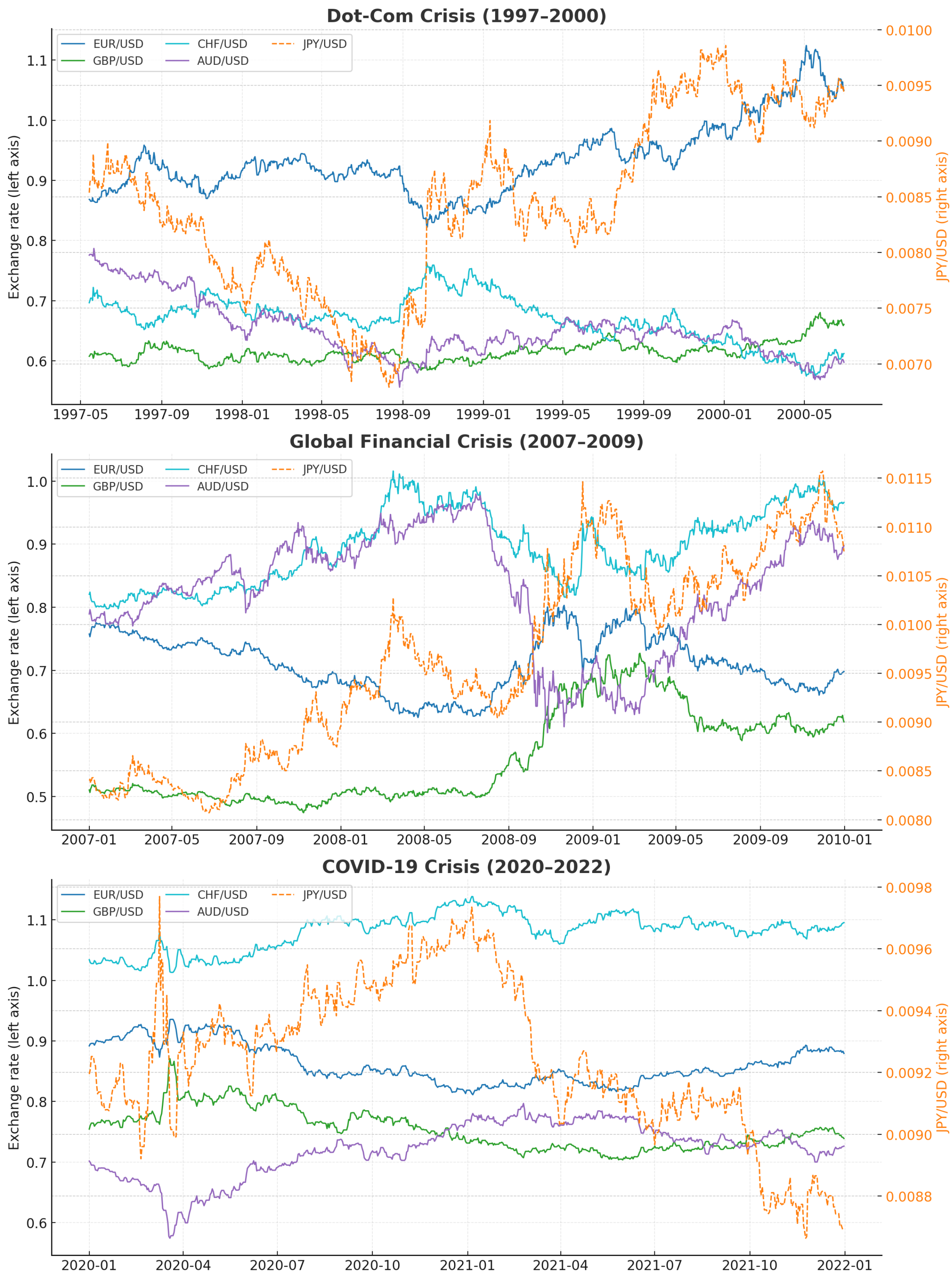

Figure 1: Spot FX movements during the Dot-Com, GFC, and COVID-19 crises.

Figure 1 illustrates the behavior of major exchange rates across the three crises. The Dot-Com period shows moderate but unsustained swings; the Global Financial Crisis features sharp volatility and pronounced safe-haven effects (e.g., CHF appreciation, AUD depreciation); and the COVID-19 crisis displays elevated volatility but with fragmented and less synchronized co-movements.

5 Methodology

5.1 Unit root tests

We begin by verifying the order of integration of each exchange rate series using the **Augmented Dickey Fuller** and **Phillips Perron** tests. Specifically:

- **Levels:** We include intercepts (and deterministic trends when appropriate) to test for unit roots and confirm nonstationarity integrated of order one.
- **First Differences:** We confirm stationarity integrated of order zero for returns.

This ensures that all series share the same integration order, a precondition for Johansen cointegration analysis. For both Augmented Dickey Fuller and Phillips Perron tests, the null hypothesis is the presence of a unit root (nonstationarity), while the alternative is stationarity. Tests are implemented in both levels and first differences, and the resulting statistics are reported below.

5.2 Johansen cointegration

Having established the integration properties of the data, we apply the **Johansen (1988, 1991)** cointegration procedure to examine whether stable long-run equilibrium relationships exist among the $N = 5$ major currencies. The vector error correction representation is:

$$\Delta Y_t = \mu + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t, \tag{15}$$

where Y_t is the $N \times 1$ vector of log spot rates, Γ_i capture short-run dynamics, and $\Pi = \alpha\beta'$ captures long-run equilibrium relations. Here, β contains the cointegrating vectors, while α are the speed-of-adjustment coefficients. If $r = \text{rank}(\Pi) = 0$, no long-run equilibrium exists; if $0 < r < N$, then r distinct cointegrating relationships characterize long-run dynamics. Johansen proposes two likelihood-ratio statistics to determine r:

The trace test, which evaluates the null hypothesis of at most r cointegrating relations against the alternative of more than r:

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{N} \ln(1 - \hat{\lambda}_i), \tag{16}$$

where T is the sample size and $\hat{\lambda}_i$ are the estimated eigenvalues (canonical correlations), ordered as $\hat{\lambda}_1 > \hat{\lambda}_2 > \cdots > \hat{\lambda}_N$.**The maximum eigenvalue test**, which evaluates the null of exactly r cointegrating relations against the alternative of $r + 1$:

$$\lambda_{\max}(r, r+1) = -T \ln(1 - \hat{\lambda}_{r+1}). \tag{17}$$

Both tests are widely used in empirical finance applications. However, Lütkepohl, Saikkonen, and Trenkler (2001) show that the trace statistic generally exhibits superior power in small and moderate samples. Since our segmented crisis-period datasets are not especially large, we therefore base most of our cointegration conclusions on the trace statistic.

5.2.1 Normalization in Johansen tests

Johansen's procedure is invariant to the choice of normalization: selecting a variable to normalize on only affects presentation of the cointegrating vector, not the rank. In our analysis, we normalize on the euro (EUR/USD). The euro is the most liquid and least policy-managed non-USD currency, making it a natural benchmark for global FX linkages. Alternative numéraires (GBP, AUD, CHF, JPY) are more prone to idiosyncratic or regime-specific shocks, which could bias interpretation of the vectors.

5.3 Vector error correction model (VECM)

When cointegration is established, we estimate a vector error correction model to quantify the strength of the equilibrium relationships. The speed of adjustment coefficients in α measure how quickly deviations from long-run equilibrium are corrected. A larger (in absolute value) coefficient signals a stronger restoring force toward equilibrium, which we interpret as evidence of robust cointegration dynamics.

5.4 Rolling return correlations

To capture broader patterns of dependence that may arise even in the absence of cointegration (i.e., without common stochastic trends binding currencies together), we compute rolling returns for each currency over horizons of one day, one week, one month, and one quarter. We then examine correlations across the five currency pairs to assess the strength of their linkages at different horizons.

5.5 Average pairwise correlations

As a parsimonious measure of the overall degree of dependence among the five currencies, we compute the simple average of all pairwise correlations of their returns. Let $R_{i,t}$ denote the return of currency i at horizon h (where $i = 1, \dots, 5$), and let ρ_{ij} denote the sample correlation coefficient between $R_{i,t}$ and $R_{j,t}$ for $i \neq j$. The total number of distinct pairwise correlations is

$$\binom{5}{2} = 10.$$

The average correlation is then defined as

$$\bar{\rho} = \frac{1}{\binom{5}{2}} \sum_{i<j} \rho_{ij},$$

which corresponds to the mean off-diagonal element of the 5×5 correlation matrix of returns.

A higher value of $\bar{\rho}$ indicates stronger average linkages among the currencies, consistent with greater market integration or contagion, while a lower value implies weaker connections and greater potential for diversification. Although this measure averages across heterogeneous bilateral relationships, it provides a tractable single-index summary of systemic currency return dependence. We track $\bar{\rho}$ across horizons and crisis periods to assess how exchange rate co-movement evolves over time.

5.5.1 Complementarity of approaches

While cointegration analysis and correlation measures may appear similar in spirit, they capture distinct aspects of exchange rate dependence. Cointegration tests whether currencies share common stochastic trends, reflecting long-run equilibrium linkages that are stable over time. In contrast, correlations measure the strength of short-run co-movements in returns across different horizons, regardless of whether such movements are anchored by an equilibrium relation. It is therefore possible for currencies to be highly correlated on average without being cointegrated, or conversely, to exhibit cointegration without strong short-run correlation. By employing both approaches, we obtain a richer characterization of exchange rate interdependence that distinguishes structural long-run integration from episodic short-run dependence, particularly across different crisis regimes.

6 Results

We now turn to the empirical analysis, which directly tests the implications of our model. Proposition 2 states that if currencies load similarly on global common factors (i.e. they are cointegrated) and crisis volatility is elevated, then long-horizon return correlations must converge toward unity. By logical contrapositive, if correlations do not approach one ($\neg C$), then not both conditions can hold. Since crisis volatility almost always rises, the implication simplifies to the idea that if correlations do not become high enough to have magnitudes around one, then the currencies do not load similarly on the common factors (i.e. cointegration is destroyed and autonomy elevated). The logical derivation of this implication is set out in the Appendix.

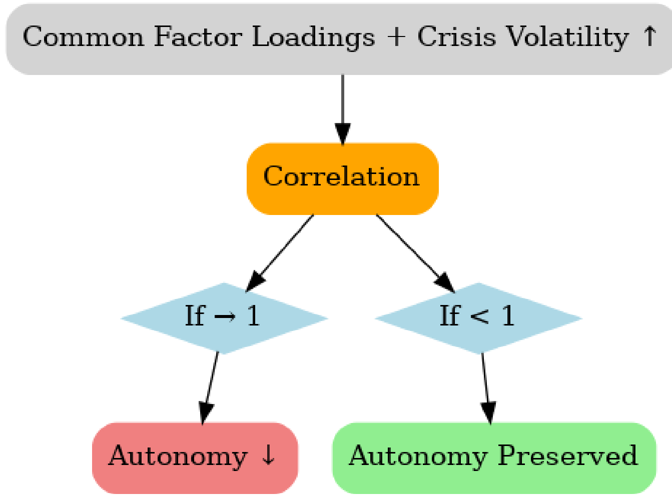

Figure 2: Stylized flow of Proposition 2 and its contrapositive. Higher crisis volatility *with* similar loadings $\Rightarrow$ correlations $\to 1 \Rightarrow$ autonomy falls; if correlations stay < 1, autonomy is preserved.

This logic provides a clear empirical test. If correlations remain modest so that we do not observe long-horizon correlations that approach one, then we conclude that autonomy persists and cointegration weakens. We evaluate these predictions using daily spot exchange rates across the three crisis subperiods. The results are presented in three stages. First, we establish the time-series properties of the data with unit root tests to confirm suitability for cointegration analysis. Second, we apply Johansen's procedure to examine the existence and strength of long-run cointegration relationships among the USD currency pairs during each crisis episode, estimating vector error correction models (where we find cointegration) to characterize the adjustment dynamics. Finally, we complement this analysis with rolling long-horizon return correlations, which provide a non-cointegration-based measure of systemic dependence and allow us to assess whether observed correlations approach the levels predicted by the model under crisis regimes. Together, these results provide an empirical test of the model's implication regarding crisis-driven convergence versus persistent currency autonomy.

Table 1: Unit root tests (Dot-Com episode)

	ADF Test		PP Test	
Variable	**t-Stat**	**p-val**	**Adj. t-Stat**	**p-val**
EUR/USD	-0.938263	0.7762	-0.903896	0.7873
Δ(EUR/USD)	-7.386294***	0.0000	-34.05348***	0.0000
JPY/USD	-1.141525	0.7011	-1.204249	0.6746
Δ(JPY/USD)	-15.27350***	0.0000	-31.62116***	0.0000
GBP/USD	-1.999517	0.2872	-1.893400	0.3120
Δ(GBP/USD)	-8.889021***	0.0000	-7.234940***	0.0000
CHF/USD	-1.385773	0.5906	-1.330779	0.6170
Δ(CHF/USD)	-23.15413***	0.0000	-34.46922***	0.0000
AUD/USD	-2.542213	0.1058	-2.477434	0.1213
Δ(AUD/USD)	-18.43708***	0.0000	-36.28420***	0.0000

Notes: ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 2: Unit root tests (Global Financial Crisis episode)

	ADF Test		PP Test	
Variable	**t-Stat**	**p-val**	**Adj. t-Stat**	**p-val**
EUR/USD	-1.756445	0.4025	-1.821258	0.3703
Δ(EUR/USD)	-32.02499***	0.0000	-32.02657***	0.0000
JPY/USD	-1.061106	0.7328	-1.081701	0.7249
Δ(JPY/USD)	-18.10110***	0.0000	-36.10449***	0.0000
GBP/USD	-0.989019	0.7590	-0.938537	0.7761
Δ(GBP/USD)	-29.92993***	0.0000	-29.84658***	0.0000
CHF/USD	-1.614304	0.4749	-1.574312	0.4954
Δ(CHF/USD)	-33.79487***	0.0000	-33.82532***	0.0000
AUD/USD	-1.248617	0.6550	-1.317173	0.6235
Δ(AUD/USD)	-35.55946***	0.0000	-35.50432***	0.0000

Notes: ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Unit root tests (COVID-19 episode)

	ADF Test		PP Test	
Variable	**t-Stat**	**p-val**	**Adj. t-Stat**	**p-val**
EUR/USD	-1.519609	0.5233	-1.680846	0.4406
Δ(EUR/USD)	-10.66765***	0.0000	-24.81385***	0.0000
JPY/USD	-0.750186	0.8316	-1.142213	0.7006
Δ(JPY/USD)	-10.81124***	0.0000	-28.37586***	0.0000
GBP/USD	-1.463991	0.5515	-1.750368	0.4054
Δ(GBP/USD)	-10.32505***	0.0000	-22.62324***	0.0000
CHF/USD	-1.988275	0.2921	-2.173474	0.2164
Δ(CHF/USD)	-10.40580***	0.0000	-25.50979***	0.0000
AUD/USD	-1.676315	0.4429	-1.578366	0.4932
Δ(AUD/USD)	-15.93793***	0.0000	-23.58878***	0.0000

Notes: ***, **, * denote significance a t the 1%, 5%, and 10% levels, respectively.

Table 4: Unrestricted cointegration rank test (Trace) — Dot Com

Hypothesized No. of CE(s)	**Eigenvalue**	**Trace Statistic**	**Critical Value (5%)**	**Prob.**
None	0.020742	49.23901	76.97277	0.8738
At most 1	0.013552	30.12296	54.07904	0.9070
At most 2	0.009714	17.67890	35.19275	0.8560
At most 3	0.007319	8.77615	20.26184	0.7570
At most 4	0.002275	2.077047	9.164546	0.7623

Notes: Prob. column reports test p-values. Stars (*, **, ***) denote 10%, 5%, 1% significance where shown.

Table 5: Unrestricted cointegration rank test (Max–eigenvalue) — Dot Com

Hypothesized No. of CE(s)	**Eigenvalue**	**Max–Eigen Statistic**	**Critical Value (5%)**	**Prob.**
None	0.020742	19.11606	34.80587	0.8639
At most 1	0.013552	12.44406	28.58808	0.9484
At most 2	0.009714	8.902748	22.29962	0.9096
At most 3	0.007319	6.699102	15.89210	0.7061
At most 4	0.002275	2.077047	9.164546	0.7623

Table 6: Unrestricted cointegration rank test (Trace) — Global Financial Crisis

Hypothesized No. of CE(s)	**Eigenvalue**	**Trace Statistic**	**Critical Value (5%)**	**Prob.**
None	0.031362	72.16038**	69.81889	0.0321
At most 1	0.020190	37.36500	47.85613	0.3305
At most 2	0.010765	15.09216	29.79707	0.7741
At most 3	0.002095	3.272948	15.49471	0.9531
At most 4	0.000899	0.982479	3.841465	0.3216

Notes: ** indicates rejection at the 5% level.

Table 7: Unrestricted cointegration rank test (Max–eigenvalue) — Global Financial Crisis

Hypothesized No. of CE(s)	**Eigenvalue**	**Max–Eigen Statistic**	**Critical Value (5%)**	**Prob.**
None	0.031362	34.79538**	33.87687	0.0388
At most 1	0.020190	22.27284	27.58434	0.2067
At most 2	0.010765	11.81921	21.13162	0.5656
At most 3	0.002095	2.290468	14.26460	0.9825
At most 4	0.000899	0.982479	3.841465	0.3216

Table 8: Unrestricted cointegration rank test (Trace) — Covid-19

Hypothesized No. of CE(s)	**Eigenvalue**	**Trace Statistic**	**Critical Value (5%)**	**Prob.**
None	0.037174	58.57631	69.81889	0.2817
At most 1	0.017760	27.51236	47.85613	0.8342
At most 2	0.010978	12.81844	29.79707	0.8997
At most 3	0.004213	3.766664	15.49471	0.9214
At most 4	0.000371	0.304418	3.841465	0.5811

Table 9: Unrestricted cointegration rank test (Max–eigenvalue) — Covid-19

Hypothesized No. of CE(s)	**Eigenvalue**	**Max–Eigen Statistic**	**Critical Value (5%)**	**Prob.**
None	0.037174	31.06395	33.87687	0.1045
At most 1	0.017760	14.69392	27.58434	0.7720
At most 2	0.010978	9.051779	21.13162	0.8282
At most 3	0.004213	3.462246	14.26460	0.9113
At most 4	0.000371	0.304418	3.841465	0.5811

Where we find sufficient evidence for cointegration, we study the dynamics as below.

6.0.1 Error–correction representation (Global Financial Crisis episode, $r = 1$)

With one cointegrating vector during the Global Financial Crisis, we normalize on EUR/USD and write

$$u_t = \mathrm{EUR/USD}_t - \beta_0 - \beta_1\,\mathrm{AUD/USD}_t - \beta_2\,\mathrm{CHF/USD}_t \\ - \beta_3\,\mathrm{GBP/USD}_t - \beta_4\,\mathrm{JPY/USD}_t, \tag{18}$$

where u_t is the equilibrium error. In long–run equilibrium $u_t = 0$; away from equilibrium $u_t \neq 0$ captures over/undervaluation of the normalized rate relative to the linear combination of the others. Let $\mathbf{s}_t = (\mathrm{EUR}, \mathrm{AUD}, \mathrm{CHF}, \mathrm{GBP}, \mathrm{JPY})'_t$ denote the five USD exchange rates and let p be the lag length in the underlying vector autoregression. The vector error–correction model is

$$\Delta\mathbf{s}_t = \boldsymbol{\alpha}\,u_{t-1} + \sum_{j=1}^{p-1} \boldsymbol{\Gamma}_j\,\Delta\mathbf{s}_{t-j} + \boldsymbol{\delta} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \text{i.i.d.}(0, \boldsymbol{\Sigma}), \tag{19}$$

with $\boldsymbol{\alpha} = (\alpha_{\mathrm{EUR}}, \alpha_{\mathrm{AUD}}, \alpha_{\mathrm{CHF}}, \alpha_{\mathrm{GBP}}, \alpha_{\mathrm{JPY}})'$ the (scalar) speeds of adjustment and u_{t-1} the lagged error–correction term. Expanding equation–by–equation gives the five short–run regressions:

$$\Delta\mathrm{EUR}_t = \alpha_{\mathrm{EUR}}\,u_{t-1} + \sum_{j=1}^{p-1} \Big(\gamma^{\mathrm{EUR}}_{\mathrm{EUR},j}\,\Delta\mathrm{EUR}_{t-j} + \gamma^{\mathrm{EUR}}_{\mathrm{AUD},j}\,\Delta\mathrm{AUD}_{t-j} + \gamma^{\mathrm{EUR}}_{\mathrm{CHF},j}\,\Delta\mathrm{CHF}_{t-j} \\ + \gamma^{\mathrm{EUR}}_{\mathrm{GBP},j}\,\Delta\mathrm{GBP}_{t-j} + \gamma^{\mathrm{EUR}}_{\mathrm{JPY},j}\,\Delta\mathrm{JPY}_{t-j}\Big) + \delta_{\mathrm{EUR}} + \varepsilon_{\mathrm{EUR},t}, \tag{20}$$

$$\Delta\mathrm{AUD}_t = \alpha_{\mathrm{AUD}}\,u_{t-1} + \sum_{j=1}^{p-1} \Big(\gamma^{\mathrm{AUD}}_{\mathrm{EUR},j}\,\Delta\mathrm{EUR}_{t-j} + \gamma^{\mathrm{AUD}}_{\mathrm{AUD},j}\,\Delta\mathrm{AUD}_{t-j} + \gamma^{\mathrm{AUD}}_{\mathrm{CHF},j}\,\Delta\mathrm{CHF}_{t-j} \\ + \gamma^{\mathrm{AUD}}_{\mathrm{GBP},j}\,\Delta\mathrm{GBP}_{t-j} + \gamma^{\mathrm{AUD}}_{\mathrm{JPY},j}\,\Delta\mathrm{JPY}_{t-j}\Big) + \delta_{\mathrm{AUD}} + \varepsilon_{\mathrm{AUD},t}, \tag{21}$$

$$\Delta\mathrm{CHF}_t = \alpha_{\mathrm{CHF}}\,u_{t-1} + \sum_{j=1}^{p-1} \Big(\gamma^{\mathrm{CHF}}_{\mathrm{EUR},j}\,\Delta\mathrm{EUR}_{t-j} + \gamma^{\mathrm{CHF}}_{\mathrm{AUD},j}\,\Delta\mathrm{AUD}_{t-j} + \gamma^{\mathrm{CHF}}_{\mathrm{CHF},j}\,\Delta\mathrm{CHF}_{t-j} \\ + \gamma^{\mathrm{CHF}}_{\mathrm{GBP},j}\,\Delta\mathrm{GBP}_{t-j} + \gamma^{\mathrm{CHF}}_{\mathrm{JPY},j}\,\Delta\mathrm{JPY}_{t-j}\Big) + \delta_{\mathrm{CHF}} + \varepsilon_{\mathrm{CHF},t}, \tag{22}$$

$$\Delta\mathrm{GBP}_t = \alpha_{\mathrm{GBP}}\,u_{t-1} + \sum_{j=1}^{p-1} \Big(\gamma^{\mathrm{GBP}}_{\mathrm{EUR},j}\,\Delta\mathrm{EUR}_{t-j} + \gamma^{\mathrm{GBP}}_{\mathrm{AUD},j}\,\Delta\mathrm{AUD}_{t-j} + \gamma^{\mathrm{GBP}}_{\mathrm{CHF},j}\,\Delta\mathrm{CHF}_{t-j} \\ + \gamma^{\mathrm{GBP}}_{\mathrm{GBP},j}\,\Delta\mathrm{GBP}_{t-j} + \gamma^{\mathrm{GBP}}_{\mathrm{JPY},j}\,\Delta\mathrm{JPY}_{t-j}\Big) + \delta_{\mathrm{GBP}} + \varepsilon_{\mathrm{GBP},t}, \tag{23}$$

$$\Delta\mathrm{JPY}_t = \alpha_{\mathrm{JPY}}\,u_{t-1} + \sum_{j=1}^{p-1} \Big(\gamma^{\mathrm{JPY}}_{\mathrm{EUR},j}\,\Delta\mathrm{EUR}_{t-j} + \gamma^{\mathrm{JPY}}_{\mathrm{AUD},j}\,\Delta\mathrm{AUD}_{t-j} + \gamma^{\mathrm{JPY}}_{\mathrm{CHF},j}\,\Delta\mathrm{CHF}_{t-j} \\ + \gamma^{\mathrm{JPY}}_{\mathrm{GBP},j}\,\Delta\mathrm{GBP}_{t-j} + \gamma^{\mathrm{JPY}}_{\mathrm{JPY},j}\,\Delta\mathrm{JPY}_{t-j}\Big) + \delta_{\mathrm{JPY}} + \varepsilon_{\mathrm{JPY},t}. \tag{24}$$

The coefficients α_i are the *speeds of adjustment*. They must be statistically significant and of signs consistent with error–correction to ensure stability. With the above normalization,

$$u_{t-1} > 0 \quad \Longrightarrow \quad \text{EUR is "too high" and/or (AUD, CHF, GBP, JPY) are "too low."}$$

Restoration of equilibrium then requires either

$$\Delta \text{EUR}_t < 0 \quad (\alpha_{\text{EUR}} < 0)$$

and/or

$$\Delta \text{AUD}_t,\ \Delta \text{CHF}_t,\ \Delta \text{GBP}_t,\ \Delta \text{JPY}_t > 0 \quad (\alpha_{\text{AUD}},\ \alpha_{\text{CHF}},\ \alpha_{\text{GBP}},\ \alpha_{\text{JPY}} > 0).$$

If a given α_i is (statistically) zero, currency i is weakly exogenous with respect to the long–run relation: it does not bear the burden of adjustment, while the currencies with significant α's do. The matrices $\mathbf{\Gamma}_j$ capture short–run propagation via lagged differences, and $\boldsymbol{\delta}$ collects deterministic terms (e.g., an unrestricted constant) as appropriate to the chosen specification. We chose an appropriate lag of $p = 2$, following the standard information criterion and the lag selection test, which was the lag used to perform the Johanssen test earlier described. Below, we present the long run equlibrium relationship and the short run dynamics.

Table 10: Johansen cointegrating vector (Global Financial Crisis; $r = 1$), normalized on EUR/USD

Variable	**Coefficient**	**Std. Error**	**t-Statistic**
EUR/USD	1.000000		
AUD/USD	-0.834022	0.20812	-4.00746
CHF/USD	-1.686385	0.24968	-6.75420
GBP/USD	-0.374197	0.22057	-1.69649
JPY/USD	0.008727	0.00224	3.90429
Constant	1.161624		

Notes: Vector is normalized such that the EUR/USD loading equals 1. Standard errors and t-statistics are reported for the non-normalized coefficients.

Table 11: Vector Error Correction Model short-run dynamics (Global Financial Crisis; dependent variables in columns)

	Dependent Variable				
	ΔEUR/USD	**ΔAUD/USD**	**ΔCHF/USD**	**ΔGBP/USD**	**ΔJPY/USD**
u_{t-1}	0.030***	−0.016	0.050***	0.020***	0.689
	(0.007)	(0.012)	(0.011)	(0.006)	(1.061)
	[4.561]	[−1.304]	[4.632]	[3.355]	[0.656]
ΔEUR_{t-1}	0.189**	−0.216	0.245*	0.042	3.647
	(0.0916)	(0.1684)	(0.1473)	(0.0817)	(14.4895)
	[2.060]	[−1.284]	[1.666]	[0.510]	[0.252]
ΔEUR_{t-2}	−0.197**	0.295*	−0.194	−0.129	14.616
	(0.0905)	(0.1663)	(0.1455)	(0.0807)	(14.3113)
	[−2.174]	[1.771]	[−1.336]	[−1.599]	[1.021]
ΔAUD_{t-1}	0.013	−0.168 ***	0.006	0.031	−10.751**
	(0.0332)	(0.0611)	(0.0534)	(0.0296)	(5.2553)
	[0.389]	[−2.750]	[0.121]	[1.048]	[−2.046]
ΔAUD_{t-2}	0.044	−0.072	0.059	0.022	−8.853*
	(0.0332)	(0.0611)	(0.0534)	(0.0296)	(5.2543)
	[1.326]	[−1.185]	[1.106]	[0.736]	[−1.685]
ΔCHF_{t-1}	−0.077	−0.000	−0.108	−0.026	−6.383
	(0.0500)	(0.0920)	(0.0805)	(0.0446)	(7.9141)
	[−1.544]	[−0.002]	[−1.347]	[−0.578]	[−0.807]
ΔCHF_{t-2}	0.087*	−0.235**	0.077	0.064	−16.633**
	(0.0499)	(0.0917)	(0.0802)	(0.0445)	(7.8895)
	[1.746]	[−2.567]	[0.958]	[1.438]	[−2.108]
ΔGBP_{t-1}	−0.065	0.057	0.018	0.110**	11.921
	(0.0573)	(0.1054)	(0.0922)	(0.0511)	(9.0658)
	[−1.135]	[0.546]	[0.195]	[2.156]	[1.315]
ΔGBP_{t-2}	0.150***	−0.195*	0.211**	0.083*	−17.228*
	(0.0568)	(0.1045)	(0.0914)	(0.0507)	(8.9923)
	[2.641]	[−1.869]	[2.308]	[1.635]	[−1.916]
ΔJPY_{t-1}	−0.000033	0.000918	−0.000554	−0.000668**	−0.02286
	(0.00033)	(0.00061)	(0.00053)	(0.00029)	(0.05208)
	[−0.101]	[1.517]	[−1.047]	[−2.276]	[−0.439]
ΔJPY_{t-2}	−0.000544*	0.001989***	−0.000329	−0.000608**	0.1652***
	(0.00033)	(0.00061)	(0.00053)	(0.00030)	(0.05257)
	[−1.637]	[3.257]	[−0.615]	[−2.053]	[3.142]
Constant	−0.0000456	−0.0000129	−0.000225	0.000103	−0.03258
	(0.00019)	(0.00035)	(0.00030)	(0.00017)	(0.02995)
	[−0.241]	[−0.037]	[−0.740]	[0.610]	[−1.088]
R^2	0.0519	0.0312	0.0584	0.0511	0.0536

Notes: Entries are coefficients with (standard errors) and [t-statistics]. u_{t-1} is the lagged error-correction term from Table 10. Asterisks (***) indicate statistical significance at 1%.

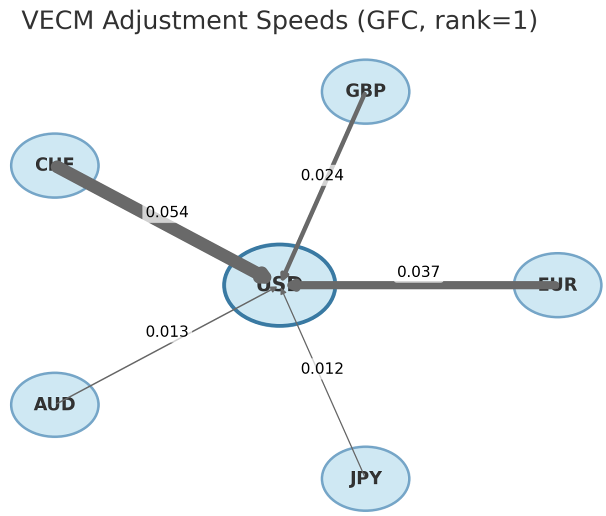

Figure 3: Vector Error Correction Model adjustment speeds during the Global Financial Crisis ($r = 1$). Arrow thickness reflects the magnitude of each estimated adjustment coefficient α_i.

Figure 3 shows the estimated error–correction speeds (α_i) from the vector error correction model during the GFC episode. EUR, CHF, and GBP have statistically significant coefficients of roughly 0.03, 0.05, and 0.02, indicating that these currencies bear the main burden of restoring the long-run equilibrium. AUD and JPY, by contrast, are insignificant. The yen's coefficient is numerically large (0.689) but highly imprecise, consistent with its weakly exogenous behavior and safe-haven status; the diagram therefore represents AUD and JPY with thin arrows. Overall, the visualization confirms the regression evidence: adjustment was concentrated in the European majors, while AUD and especially JPY remained largely outside the error–correction mechanism.

6.0.2 Cointegration and adjustment (GFC vs. Dot-Com/Covid-19)

During the Global Financial Crisis we find one cointegrating relation among USD quotes of the majors. The long-run combination loads positively on AUD/CHF/GBP and (slightly) negatively on JPY (up to normalization), and deviations are corrected primarily through EUR, CHF and GBP with daily half-lives of roughly 2–5 weeks. Dot-Com and Covid-19 show no cointegration. These results align with our theory: in systemic USD-centered stress (Global Financial Crisis) autonomy compresses, co-movement strengthens, and a stable anchor emerges; when shocks are more heterogeneous (Dot-Com, Covid-19), autonomy rises and the long-run link disappears.[5]

[5] With the vector normalized on EUR/USD, estimated coefficients are $\{\beta_{\mathrm{AUD}}, \beta_{\mathrm{CHF}}, \beta_{\mathrm{GBP}}, \beta_{\mathrm{JPY}}\} = \{-0.834, -1.686, -0.374, 0.009\}$. Multiplying the whole vector by -1 gives the equivalent "positive loadings

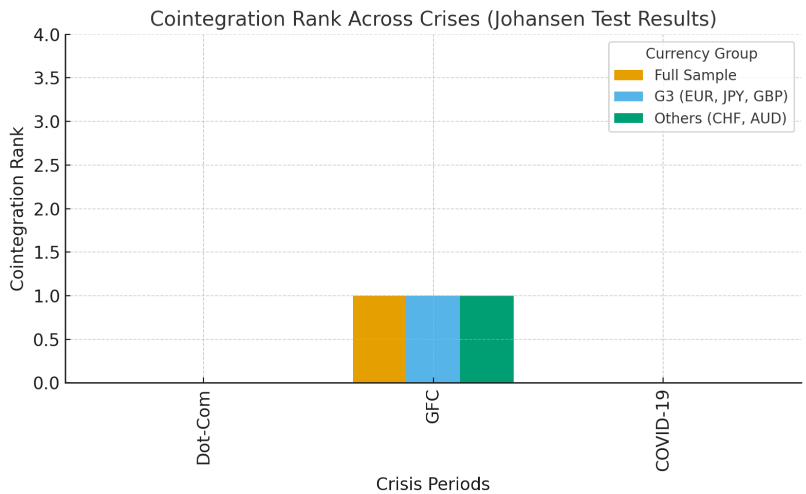

Figure 4: Cointegration rank across crises based on Johansen trace test results. The Dot-Com and COVID-19 episodes exhibit no cointegration (rank = 0), whereas the Global Financial Crisis yields one cointegrating relation (rank = 1). Bars for the full sample and subgroups (G3 vs. others) illustrate the robustness of this finding.

6.0.3 Error-correction (daily speeds)

The Error Correction Model coefficients on u_{t-1} are:

$$\alpha_{\text{EUR}} = 0.030^{***}, \alpha_{\text{CHF}} = 0.050^{***}, \alpha_{\text{GBP}} = 0.020^{***}, \alpha_{\text{AUD}} = -0.016\,(\text{ns}), \alpha_{\text{JPY}} = 0.689\,(\text{ns}).$$

EUR, CHF, and GBP adjust significantly to deviations from the long-run relation; AUD and especially JPY do not. In the Global Financial Crisis, the long-run parity is therefore restored primarily by EUR/CHF/GBP, with JPY behaving as weakly exogenous to the cointegrating space (consistent with its safe-haven role).

Half-lives. Using $HL \approx \ln(2)/\alpha$ at a daily frequency: CHF $\approx$ 14 days ($\alpha \approx 0.05$), EUR $\approx$ 23 days ($\alpha \approx 0.03$), and GBP $\approx$ 35 days ($\alpha \approx 0.02$). These magnitudes are reasonable for daily foreign exchange.

6.0.4 Link to the propositions

The Global Financial Crisis results are a clean instance of autonomy compression: a strong common USD factor and similar (non-negligible) loadings across most majors yield (i) a detectable cointegrating anchor and (ii) faster error-correction in the non-yen pairs—exactly what our proposition predicts when global factor volatility rises and currency responses

on AUD/CHF/GBP, slight negative on JPY" phrasing; the economic content is invariant to this normalization.

align. By contrast, Dot-Com and Covid-19 show no cointegration, fitting the heterogeneous response or higher autonomy side of the narrative. We now turn to the short-run dynamics. Table 11 reports the vector error correction model results for daily returns during the Global Financial Crisis. The lagged error-correction term u_{t-1} is strongly significant in the EUR, CHF, and GBP equations ($t = 4.56$, 4.63, and 3.36, respectively), but insignificant for AUD and JPY. Thus, in the short run, deviations from the unique long-run relation are corrected primarily through the European majors, while AUD and (especially) JPY behave as weakly exogenous to the cointegrating space. The magnitudes imply economically fast convergence at a daily frequency: estimated half-life is approximately two weeks for CHF ($|\alpha| \approx 0.05$), three weeks for EUR ($|\alpha| \approx 0.03$), and five weeks for GBP ($|\alpha| \approx 0.02$).[6]

Own-lag dynamics are intuitive. EUR shows one-day momentum followed by two-day reversal ($\Delta\mathrm{EUR}_{t-1} > 0$, significant; $\Delta\mathrm{EUR}_{t-2} < 0$, significant). AUD mean-reverts ($\Delta\mathrm{AUD}_{t-1} < 0$), while JPY exhibits two-day persistence ($\Delta\mathrm{JPY}_{t-2} > 0$). GBP displays short-run momentum ($\Delta\mathrm{GBP}_{t-1} > 0$).

Cross-currency spillovers align with standard risk/safe-haven channels. A stronger CHF two days ago predicts weaker AUD today ($\Delta\mathrm{CHF}_{t-2} < 0$ in the AUD equation), consistent with safe-haven strength foreshadowing a retreat in a high-beta currency. Past risk-on moves in AUD precede subsequent JPY underperformance ($\Delta\mathrm{AUD}_{t-1} < 0$ in the JPY equation), consistent with the yen's safe-haven profile. Sterling transmits to continental Europe with a short delay: $\Delta\mathrm{GBP}_{t-2}$ significantly raises both $\Delta\mathrm{EUR}$ and $\Delta\mathrm{CHF}$. Conversely, JPY pressure spills into GBP, with both $\Delta\mathrm{JPY}_{t-1}$ and $\Delta\mathrm{JPY}_{t-2}$ negative in the GBP equation.

Unsurprisingly for daily FX data, the R^2 values are modest (3–6%), but the sign and significance patterns are clear: (i) short-run adjustment to long-run parity is borne by EUR/CHF/GBP; (ii) AUD acts as a high "risk-beta" currency—mean-reverting on its own and responding to safe-haven flows; and (iii) JPY remains largely outside the error-correction margin.

This concentration of adjustment on non-yen European majors, with AUD and JPY playing distinct risk and safe-haven roles, dovetails with our regime-based interpretation. During globally systemic, USD-centered stress—such as the GFC—a stable long-run anchor emerges, and short-run re-equilibration is executed by the currencies most tightly linked to that anchor. In contrast, the yen remains comparatively autonomous.

There was no long-run equilibrium relationship to adjust to (i.e., no cointegration) during either the Dot-Com or COVID-19 stress episodes. While the Dot-Com crisis was USD-centered, it was not globally systemic. Conversely, COVID-19 was globally systemic, but

[6] Half-life $\approx \ln(0.5)/\ln(1-|\alpha|)$. The sign of α depends on the normalization of the cointegrating vector; economic "speed" is governed by $|\alpha|$.

not USD-centered. Our results suggest that for a crisis to coincide with cointegrating relationships among major dollar foreign exchange pairs, it must be both globally systemic and USD-centered; one condition alone is not sufficient. The Global Financial Crisis crisis episode satisfies both conditions as it was not only globally systemic but it also originated from the US, making it USD centered.

Our evidence is consistent with (though cannot prove from three episodes alone) the idea that cointegrating links among major USD pairs tend to appear when stress is both globally systemic and USD-centered—as in the Global Financial Crisis.

6.1 Correlation of (long-horizon) rolling returns

Table 12: Dot-Com: Average Correlation of H-day-ahead foreign exchange returns (in USD)

	EUR/USD	JPY/USD	GBP/USD	CHF/USD	AUD/USD
$H = 1$ day ahead	-0.20	0.02	-0.09	-0.25	0.01
$H = 7$ days ahead	-0.19	0.05	-0.10	-0.26	0.02
$H = 30$ days ahead	-0.25	-0.03	-0.18	-0.25	-0.03
$H = 90$ days ahead	-0.26	-0.04	-0.17	-0.22	0.02
$H = 180$ days ahead	-0.13	0.02	-0.11	-0.29	0.13
$H = 365$ days ahead	0.12	0.34	0.23	-0.56	0.31

Table 13: GFC: Average Correlation of H-day-ahead foreign exchange returns (in USD)

	EUR/USD	JPY/USD	GBP/USD	CHF/USD	AUD/USD
$H = 1$ day ahead	-0.24	0.00	-0.08	-0.13	-0.30
$H = 7$ days ahead	-0.25	0.01	-0.07	-0.14	-0.30
$H = 30$ days ahead	-0.28	0.05	-0.10	-0.11	-0.28
$H = 90$ days ahead	-0.24	0.09	-0.06	-0.15	-0.34
$H = 180$ days ahead	-0.20	0.07	-0.08	-0.22	-0.36
$H = 365$ days ahead	-0.26	0.02	-0.20	-0.20	-0.29

The average correlations of H-horizon currency returns, where foreign exchange is expressed as the value of 1 unit currency in USD, are shown below in Table 12, 13, and 14. The horizon correlations mostly fall in the weak-to-moderate range (0.1–0.4 in absolute value), often near zero and sometimes negative, and remain far from one. This indicates that currencies do not move in a tightly synchronized way in response to common USD shocks. Interpreted through Proposition 2, this pattern is consistent with higher autonomy and, accordingly, a reduced scope for long-run links, matching the lack of cointegration in Dot-Com and Covid-19. By contrast, during the GFC we recover a single cointegrating relation. That the cointegration

Table 14: Covid-19: Average Correlation of H-day-ahead FX returns (in USD)

	EUR/USD	**JPY/USD**	**GBP/USD**	**CHF/USD**	**AUD/USD**
$H = 1$ day ahead	-0.34	-0.03	-0.25	-0.08	-0.16
$H = 7$ days ahead	-0.36	-0.01	-0.28	-0.07	-0.15
$H = 30$ days ahead	-0.33	0.02	-0.20	-0.07	-0.22
$H = 90$ days ahead	-0.37	0.05	-0.16	-0.02	-0.19
$H = 180$ days ahead	-0.48	0.11	-0.20	0.08	-0.09
$H = 365$ days ahead	-0.48	0.07	-0.26	0.04	-0.07

rank, i.e. number of cointegration relations, equals $1 < 4$ (the $N-1$ maximum for $N = 5$ USD quotes) shows that meaningful cross-currency heterogeneity persists, which is why a richer set of cointegrating relations does not materialize.

A critical clarification is warranted on the distinction between visual inspection, correlation, and cointegration in our analysis. While graphical inspection of exchange rate series may offer intuition, "seeing is not believing." Most financial time series do not display relationships that can be reliably inferred by eye. Statistical measures are indispensable: to establish whether currencies co-move strongly, one must compute formal correlation measures. Without such metrics, inference risks being misleading.

Our results confirm that correlations among long-horizon currency *returns* mostly lie below 0.4 in magnitude, with the bulk of values on the weak side. Crucially, these correlations must be computed on returns rather than levels, since non-stationary levels would spuriously inflate correlation. By contrast, cointegration analysis is necessarily performed on *levels*, because long-run equilibrium concerns the presence of common stochastic trends in the underlying processes, not short-run fluctuations. Thus, correlation and cointegration are conceptually distinct: correlation measures short-run comovement in returns, while cointegration tests for shared stochastic trends in levels.[7]

This distinction underpins Proposition 2. If currencies load similarly on global stochastic trends, then long-horizon return correlations should converge toward one. The contrapositive is equally informative: if correlations do *not* converge, then loadings must be heterogeneous or additional independent stochastic trends remain. Our evidence aligns partly in the case of the Global Financial Crisis, and fully in the cases of the Dot-Com and COVID-19 crises, with this logic, consistent with the contrapositive interpretation of Proposition 2. During the **Dot-Com** and **COVID-19** crises, Johansen tests indicate no cointegration (rank = 0), implying

[7] As a simple example, two independent random walks with drift (say, EUR/USD and GBP/USD) will often show high correlations in levels purely because both trend upward over time. Their returns, however, are uncorrelated. This illustrates why correlations on levels are misleading and why cointegration is the correct tool for testing shared long-run trends.

full autonomy among currencies with no common stochastic trends. In these regimes, long-horizon correlations fail to converge to one, exactly as Proposition 2 predicts. During the **Global Financial Crisis**, by contrast, we find one cointegrating relation (rank = 1 of a possible four), implying partial commonality: some stochastic trends were shared, others idiosyncratic. Correspondingly, long-horizon correlations exhibit only partial alignment, again consistent with the proposition.

To visualize this implication, Figures 5–7 report rolling 6-month correlations of USD currency returns against EUR/USD during each crisis episode. These plots reveal whether correlations moved toward the theoretical full-convergence benchmark of +1 (Note that EUR/USD itself is the base series, so its correlation with itself is identically +1; the dotted line thus serves only as a theoretical benchmark rather than an additional data series.) under elevated volatility. In the **Dot-Com** and **COVID-19** panels, correlations remain volatile and often near zero or negative, with no line persistently approaching one. By contrast, during the **Global Financial Crisis**, correlations rise more strongly, and the EUR–GBP pair in particular stabilizes near 0.75–0.9. This distinctive elevation signals a partial but not universal alignment of currency dynamics.

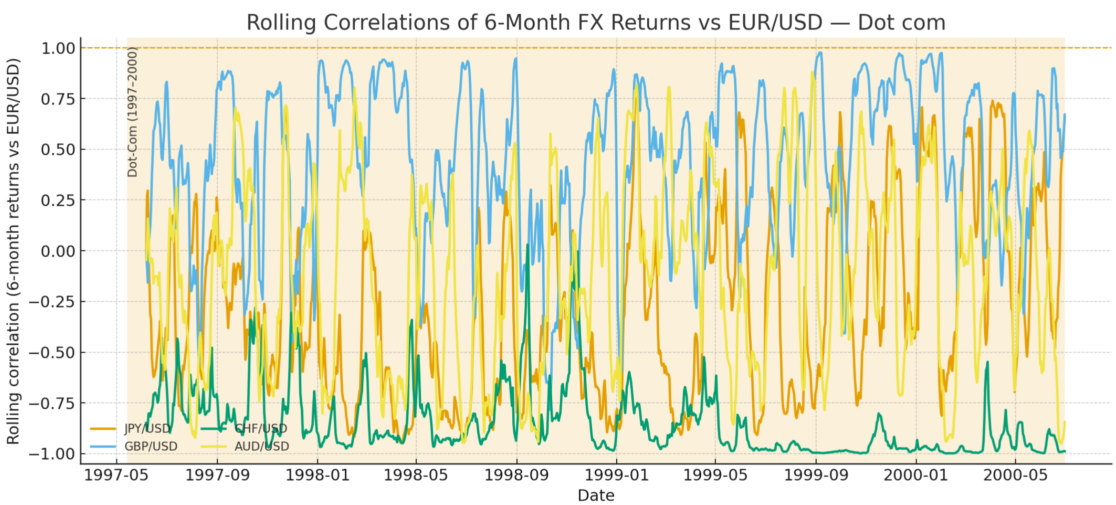

Figure 5: Dot-Com (1997–2000).

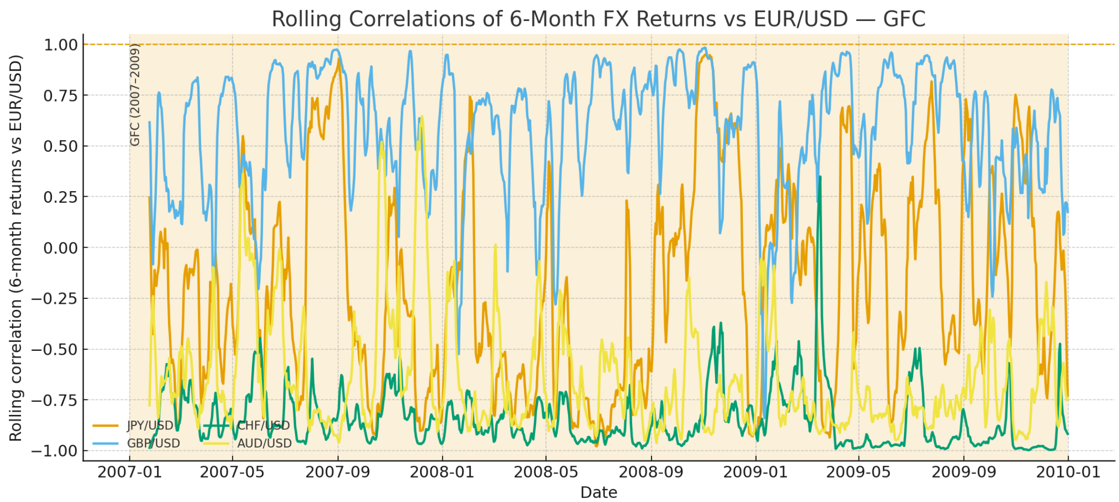

Figure 6: GFC (2007–2009).

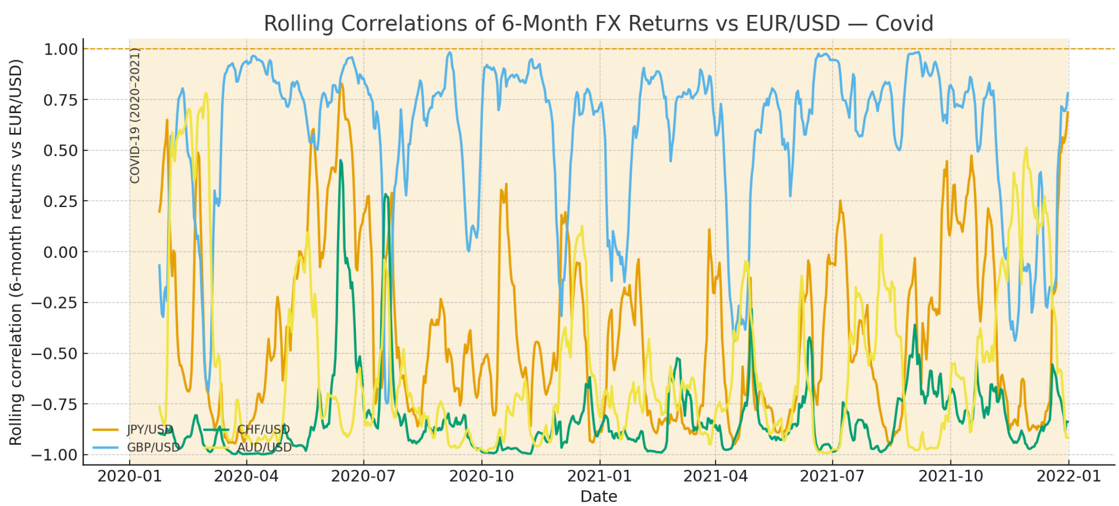

Figure 7: COVID-19 (2020–2021).

As a complementary robustness check, Figure 8 first reports the JP Morgan Global Currency Volatility Index (JPMVXYGL) with crisis windows highlighted, providing the underlying volatility benchmark. Building on this, Figure 9 then plots the same long-horizon rolling correlations against log(**JPMVXYGL**). This cross-sectional view directly tests Proposition 2: during the GFC, high volatility coincided with partial convergence, while in the Dot-Com and COVID-19 episodes, high volatility corresponded to dispersed or negative correlations, consistent with heterogeneous factor loadings and the absence of cointegration.

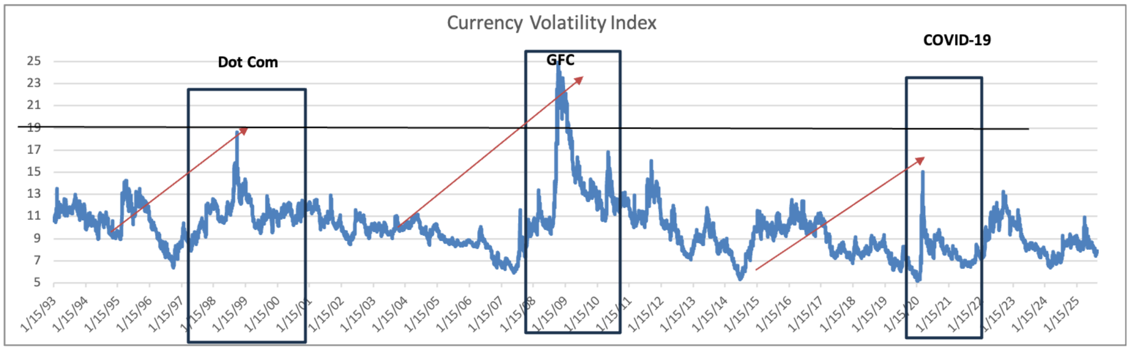

Figure 8: JP Morgan Global Currency Volatility Index with Dot-Com, Global Financial Crisis, and COVID-19 windows highlighted.

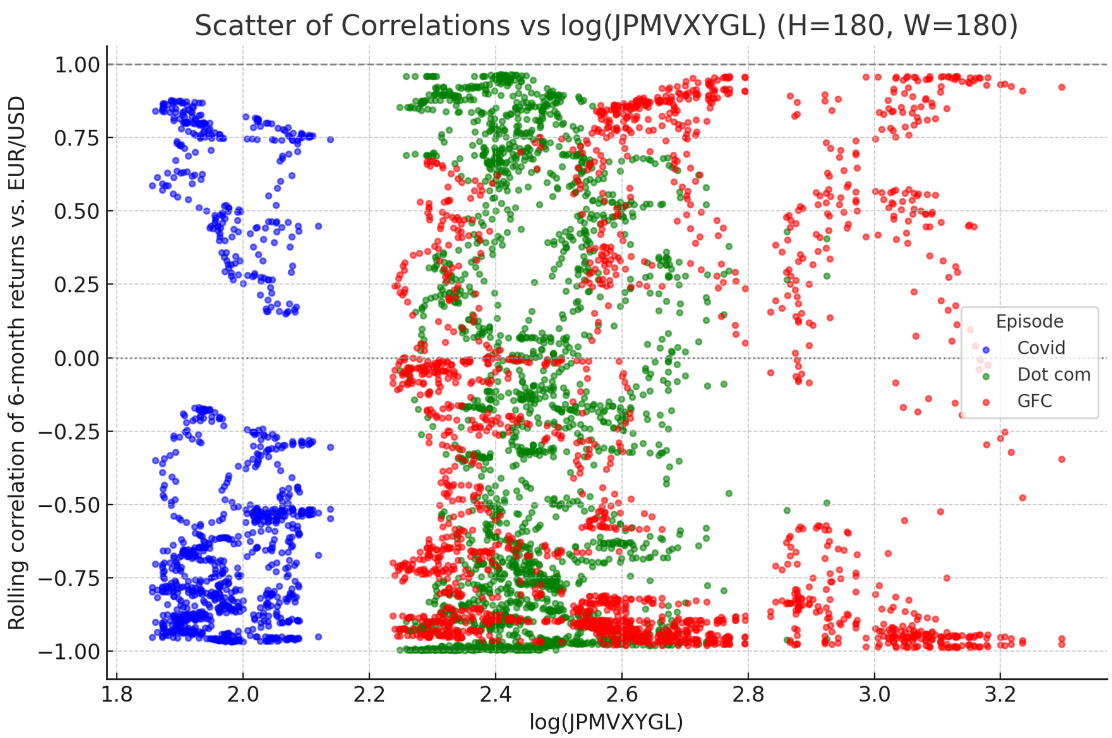

Figure 9: Scatter of rolling 6-month correlations vs. log(JPMVXYGL). Points are colored by crisis windows: red = Global Financial Crisis (2007–2009), green = Dot-Com (1997–2000), blue = COVID-19 (2020–2021). The figure shows that only in the Global Financial Crisis does high volatility coincide with partial convergence.

In sum, three points emerge: (i) correlations must be computed on returns, not levels; (ii) cointegration must be tested on levels, not returns; and (iii) correlation and cointegration are distinct and non-equivalent, with lack of convergence in correlations providing evidence of heterogeneous loadings or additional independent trends.

7 Conclusion

We show that long-run foreign exchange linkages during crisis episodes are crisis-regime dependent. Not all crisis episodes are born equal. We present a stylized model to demonstrate this. Empirically, using Johansen tests and a vector error correction model on five major USD pairs, we examine three major crisis episodes of Dot Com, Global Financial Crisis and Covid-19 and find that the Global Financial Crisis is the only episode with a surviving long-run cointegration relation (with cointegration rank = 1); Dot-Com and COVID-19 show none or much less evidence, weaker evidence, whatever. A simple factor mapping links this to the rank of the pass-through matrix: when stress is both global and USD-centered, country's autonomy compresses and a stable anchor (i.e. cointegration among variables that have lost their automomy and now forced to reach to common global factors stochastic trends) can emerge; when shocks are heterogeneous or introduce an additional persistent trend, rank of stochastic trends q_r rises and cointegration disappears with the increase in autonomy or increase in the number of stochastic trends (u can call this number "rank", let's be consistent because cointegration also has rank, you guys can laser scrutinize this). Error-correction loads on EUR/CHF/GBP as they are the ones that adjust in restoring the long run equilibrium relationship among the currencies when a deviations or break down/scattering of this stable long run relationships occurs in the short run/or briefly—with daily half-lives of 2–5 weeks—while AUD and especially JPY are weakly exogenous, consistent with risk/safe-haven roles.

The distinction between rank—the number of stochastic trends q_r (not the cointegration rank)—and tightness (correlation) explains why crises can exhibit elevated co-movement even in the absence of long-run parity. Practically, diversification across USD majors can fail in USD-centered systemic stress; absent cointegration. Our framework yields testable implications: (i) stronger common-shock intensity and more aligned loadings should coincide with lower rank and higher cointegration rank; (ii) heterogeneous or novel persistent shocks should raise rank and lower cointegration rank (cointegration count). Our paper verifies these predictions using the world's most liquid currency sets viz a viz the USD.

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8 Appendix

In the Appendix, we provide all the proofs of all technical results.

8.1 Proof of Proposition 1 (Cointegration rank $N-1$)

To prove Proposition 1, the following lemma is applied.

Lemma 2. *(Cointegration rank $N - \text{rank}(\Lambda_r)$) Given regime r and assumptions A1–A4, the cointegration rank of X_t is:*

$$\textit{Cointegration rank of } X_t = N - \text{rank}(\Lambda_r).$$

Proof. Consider the long-run factor structure:

$$X_t = \mu + \Lambda_r g_t + \nu_t,$$

where:

- $X_t \in \mathbb{R}^N$ is a vector of exchange rates (e.g., log bilateral exchange rates),
- $\Lambda_r \in \mathbb{R}^{N \times K}$ is the matrix of factor loadings,
- $g_t \in \mathbb{R}^K$ is a vector of global stochastic trends, assumed to be integrated of order one,
- ν_t is a stationary vector process.

By assumption A2, the components of g_t are mutually uncorrelated and not cointegrated. Thus, g_t represents K independent stochastic trends. The nonstationarity in X_t arises only from the term $\Lambda_r g_t$. Let $r := \mathrm{rank}(\Lambda_r) \leq \min(N, K)$. Then the image of g_t under Λ_r spans an r-dimensional subspace of $\mathbb{R}^N$. This implies that the nonstationary component of X_t lies entirely within an r-dimensional subspace. Let us now consider the existence of cointegrating vectors. A cointegrating vector $\beta \in \mathbb{R}^N$ is such that the linear combination $\beta^\top X_t$ is stationary:

$$\beta^\top X_t = \beta^\top \mu + \beta^\top \Lambda_r g_t + \beta^\top \nu_t.$$

Since μ is constant and ν_t is stationary, the stationarity of $\beta^\top X_t$ depends entirely on whether $\beta^\top \Lambda_r g_t$ is stationary. But g_t is I(1), so $\beta^\top \Lambda_r g_t$ is stationary if and only if:

$$\beta^\top \Lambda_r = 0.$$

That is, cointegrating vectors lie in the left null space of Λ_r. The dimension of this null space is:

$$\dim\left(\{\beta \in \mathbb{R}^N : \beta^\top \Lambda_r = 0\}\right) = N - \mathrm{rank}(\Lambda_r).$$

Hence, the number of linearly independent cointegrating vectors—i.e., the cointegration rank of X_t—is:

$$\text{Cointegration rank} = N - \mathrm{rank}(\Lambda_r).$$

$\square$

Proof of Proposition 1. For any $b \in \mathbb{R}^N$, note:

$$b' X_t = b' \mu_r + (b' \Lambda_r)\, g_t + b' \Gamma z_t + b' u_t.$$

Since z_t and u_t are stationary, $b'\Gamma z_t + b' u_t$ is stationary integrated of order zero. The only potentially nonstationary component is $(b'\Lambda_r) g_t$, because g_t is integrated of order one and its components are assumed not to be cointegrated (so they represent independent stochastic trends). Thus:

- If $b'\Lambda_r = 0$, then $b' X_t$ is a constant plus stationary variables $I(0)$ terms $\Rightarrow$ $I(0)$.
- If $b'\Lambda_r \neq 0$, then $b' X_t$ inherits an $I(1)$ component $\Rightarrow$ $I(1)$.

Hence, the cointegrating space is $\{b \in \mathbb{R}^N : b'\Lambda_r = 0\}$, of dimension $N - 1$ whenever $\mathrm{rank}(\Lambda_r) = 1$. This follows from Lemma 2.

8.2 Proof of Proposition 2 (Crisis Co-Movement Amplification of Long Horizon Currency Returns)

The proof of Proposition 2 is divided into three steps. We start by computing the variance of the long-run returns, then the factor share of the total variance, and conclude by computing the correlation between long-run returns.

Proof of Proposition 2. We begin with the factor model for currency i in regime r:

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^{\top} g_t + \Gamma_i^{\top} z_t + u_{i,t},$$

where g_t are global stochastic trends (e.g., integrated), z_t are stationary domestic/regional factors, and $u_{i,t}$ is idiosyncratic noise.

Taking first differences:

$$\Delta X_{i,t} = \lambda_{i,r}^{\top} \Delta g_t + \Delta(\Gamma_i^{\top} z_t + u_{i,t}).$$

Define the cumulative return over horizon H as:

$$Y_{i,H} \equiv \sum_{h=1}^{H} \Delta X_{i,t+h} = \lambda_{i,r}^{\top}(g_{t+H} - g_t) + \varepsilon_{i,H},$$

where

$$\varepsilon_{i,H} = \Gamma_i^{\top}(z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t})$$

is a difference of stationary processes.

Step 1 (Variance of the long-run return): Because g_t is a vector of random walks,

$$\mathrm{Var}(g_{t+H} - g_t) = H\Sigma_{g,r}.$$

With $\varepsilon_{i,H}$ uncorrelated with g_t, we obtain

$$\mathrm{Var}(Y_{i,H}) = H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + \mathrm{Var}(\varepsilon_{i,H}).$$

Since $\varepsilon_{i,H}$ is a *difference of stationary levels*, its variance is bounded:

$$\mathrm{Var}(\varepsilon_{i,H}) = O(1).$$

Hence

$$\mathrm{Var}(Y_{i,H}) = H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + O(1).$$

Step 2 (Factor share of total variance): The share of total variance explained by global factors at horizon H is

$$\frac{\text{Factor variance}}{\text{Total variance}} = \frac{H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r}}{H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + O(1)} = \frac{\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r}}{\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + O(1/H)}.$$

Thus, as $\Sigma_{g,r}$ rises in crisis, this ratio rises; and as $H \to \infty$, the $O(1/H)$ term vanishes so the factor share tends to 1. Thus, the common global factors explain a larger fraction of the long-run variance. This is the *variance channel* of co-movement amplification.

Step 3 (Correlation between long-run returns): For any pair i, j, define:

$$\rho_{ij}^{(H)} = \frac{\mathrm{Cov}(Y_{i,H}, Y_{j,H})}{\sqrt{\mathrm{Var}(Y_{i,H})\,\mathrm{Var}(Y_{j,H})}}.$$

We have

$$\mathrm{Cov}(Y_{i,H}, Y_{j,H}) = H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{j,r} + O(1),$$

$$\mathrm{Var}(Y_{i,H}) = H\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + O(1), \qquad \mathrm{Var}(Y_{j,H}) = H\lambda_{j,r}^{\top}\Sigma_{g,r}\lambda_{j,r} + O(1).$$

Dividing numerator and denominator by H,

$$\rho_{ij}^{(H)} = \frac{\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{j,r} + O(1/H)}{\sqrt{\big(\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r} + O(1/H)\big)\big(\lambda_{j,r}^{\top}\Sigma_{g,r}\lambda_{j,r} + O(1/H)\big)}}.$$

As $H \to \infty$, the bounded stationary differences vanish:

$$\lim_{H\to\infty} \rho_{ij}^{(H)} = \frac{\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{j,r}}{\sqrt{(\lambda_{i,r}^{\top}\Sigma_{g,r}\lambda_{i,r})(\lambda_{j,r}^{\top}\Sigma_{g,r}\lambda_{j,r})}}.$$

This is the cosine of the angle between the vectors $\Sigma_{g,r}^{1/2}\lambda_{i,r}$ and $\Sigma_{g,r}^{1/2}\lambda_{j,r}$. If the factor loadings are aligned (i.e., $\lambda_{i,r} \approx \lambda_{j,r}$), then the limit approaches 1. Thus, greater similarity in how currencies respond to global shocks leads to stronger long-run correlation. This is the *loading-alignment channel*.

8.3 Proof of Proposition 3 (Rank Change in Crisis and Cointegration)

Proof of Proposition 3. Consider the factor structure of currency returns before and during crisis regimes:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma_r z_t + u_t,$$

where:

- X_t is the $N \times 1$ vector of currency returns,
- Λ_r is the $N \times M_r$ loading matrix on persistent global factors g_t,
- g_t is a $M_r \times 1$ vector of persistent global factors (stochastic trends),
- $\Gamma_r z_t + u_t$ is a stationary component,
- The regime $r \in \{$Dot-Com, GFC, COVID, the crisis periods$\}$ indexes different crisis episode periods.

Step 1 (Relation between rank of Λ_r and cointegration): By standard results in cointegration theory (see Johansen 1991; Stock and Watson 1988), the number of common stochastic trends driving X_t equals the rank of Λ_r, denoted $\text{rank}(\Lambda_r) = K_r$, where $K_r := \text{rank}(\Lambda_r) \leq \min(N, M_r)$. The dimension of the cointegrating space, i.e., the number of stationary linear combinations of X_t, equals

$$\dim(\text{cointegrating space}) = N - K_r.$$

Step 2 (Effect of new persistent global factors): Suppose that during a crisis, new persistent global factors emerge that are independent of the pre-crisis factors. Formally, the crisis regime factor vector can be written as

$$g_t^{\text{crisis}} = \begin{bmatrix} g_t^{\text{pre}} \\ g_t^{\text{new}} \end{bmatrix},$$

where $\dim(g_t^{\text{new}}) = K_{\text{new}} > 0$, and g_t^{new} is independent of g_t^{pre}. The corresponding loading matrix expands as

$$\Lambda_{\text{crisis}} = \left[\Lambda_{\text{pre}} \quad \Lambda_{\text{new}}\right],$$

with Λ_{new} loading persistently on the new factors. Because g_t^{new} is independent, the rank of Λ_{crisis} satisfies

$$\text{rank}(\Lambda_{\text{crisis}}) \geq \text{rank}(\Lambda_{\text{pre}}).$$

Thus,

$$K_{\text{crisis}} = \text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}) = K_{\text{pre}}.$$

Step 3 (Effect on cointegration relations): Since the number of cointegrating relations equals $N - K_r$, an increase in the rank K_r reduces the cointegration dimension:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

In the extreme case where $M_{\text{crisis}} \geq N$ and Λ_{crisis} attains full row rank N,

$$K_{\text{crisis}} = N,$$

the number of cointegrating relations becomes zero, meaning no stationary linear combination exists - the system has no cointegration.

Step 4 (Effect of autonomy shifts on rank): Alternatively, even without new factors, if autonomy shifts such that the loading matrix Λ changes rank (e.g., if some currency responses become linearly independent from others), then

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

and the same conclusion on cointegration applies.

Hence, the emergence of new persistent global factors or autonomy shifts that increase the rank of Λ during crisis reduce the number of cointegrating relations, and if the rank reaches N, cointegration disappears entirely.

8.4 Proof of Lemma 1

Proof of Lemma 1. Rewrite the no-arbitrage condition as a log-normal moment. From (5) and $X_{i,t} - X_{i,t+1} = -\Delta X_{i,t+1}$,

$$1 = \mathbb{E}_t\big[\, M_{t+1}\, e^{-\Delta X_{i,t+1}} \,\big] = \mathbb{E}_t\big[\exp\big(\log M_{t+1} - \Delta X_{i,t+1}\big)\big]\,.$$

Define

$$Y_{i,t+1} \; := \; \log M_{t+1} - \Delta X_{i,t+1}.$$

By the conditions of the lemma, we know that $\log M_{t+1}$ is Gaussian and $\Delta X_{i,t+1}$ is an affine function of Gaussian shocks; hence $Y_{i,t+1}$ is Gaussian conditional on $\mathcal{F}_t$:

$$Y_{i,t+1} \mid \mathcal{F}_t \; \sim \; \mathcal{N}\big(\mu_{Y,i,t},\, \upsilon_{Y,i,t}\big).$$

Therefore

$$\mathbb{E}_t\big[e^{Y_{i,t+1}}\big] \; = \; \exp\Big(\mu_{Y,i,t} + \tfrac{1}{2}\upsilon_{Y,i,t}\Big) \; = \; 1. \tag{25}$$

We next compute $\mu_{Y,i,t}$ and $\upsilon_{Y,i,t}$ and note $a_{i,r}$. To do this, we again refer to the first assumption of the lemma and plug the expression for $\log M_{t+1}$ into $Y_{i,t+1}$:

$$Y_{i,t+1} = \underbrace{\big(-r_t - \tfrac{1}{2}\varsigma_r^2\big)}_{\text{deterministic at } t} - \xi_{t+1} \; - \; \Delta X_{i,t+1}.$$

Using

$$\Delta X_{i,t+1} = a_{i,r} + \lambda_{i,r}' \, \Delta g_{t+1} + \Gamma_i' \, \Delta z_{t+1} + \epsilon_{i,t+1},$$

we have

$$Y_{i,t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 \; - \; \xi_{t+1} \; - \; a_{i,r} - \lambda_{i,r}'\Delta g_{t+1} - \Gamma_i'\Delta z_{t+1} - \epsilon_{i,t+1}.$$

Using $\xi_{t+1} = \lambda_r'\Delta g_{t+1} + \upsilon_{t+1}$, we have

$$Y_{i,t+1} = -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r} \; - \; (\lambda_r' + \lambda_{i,r}')\Delta g_{t+1} \; - \; \Gamma_i'\Delta z_{t+1} \; - \; \upsilon_{t+1} \; - \; \epsilon_{i,t+1}.$$

Hence, conditional mean and variance are

$$\begin{aligned}
\mu_{Y,i,t} &= -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r}, \\
\upsilon_{Y,i,t} &= \mathrm{Var}_t\big((\lambda_r' + \lambda_{i,r}')\Delta g_{t+1} + \Gamma_i'\Delta z_{t+1} + \upsilon_{t+1} + \epsilon_{i,t+1}\big).
\end{aligned}$$

Using the orthogonality of blocks in the lemma, this variance decomposes additively:

$$\upsilon_{Y,i,t} = (\lambda_r + \lambda_{i,r})'\Sigma_{\Delta g,r}\,(\lambda_r + \lambda_{i,r}) \; + \; \Gamma_i'\Sigma_{\Delta z}\,\Gamma_i \; + \; \sigma_\upsilon^2 \; + \; \sigma_{\epsilon_i}^2, \tag{26}$$

where $\Sigma_{\Delta g,r} = \mathrm{Var}_t(\Delta g_{t+1})$ (regime r), $\Sigma_{\Delta z} = \mathrm{Var}_t(\Delta z_{t+1})$, and $\sigma_\upsilon^2 = \mathrm{Var}_t(\upsilon_{t+1})$, $\sigma_{\epsilon_i}^2 = \mathrm{Var}_t(\epsilon_{i,t+1})$.

Imposing (25) gives

$$\mu_{Y,i,t} + \tfrac{1}{2}\upsilon_{Y,i,t} \ = \ 0 \quad \Longrightarrow \quad -r_t - \tfrac{1}{2}\varsigma_r^2 - a_{i,r} + \tfrac{1}{2}\upsilon_{Y,i,t} \ = \ 0.$$

Therefore the drift in the difference equation is pinned down by

$$a_{i,r} = -r_t - \tfrac{1}{2}\varsigma_r^2 + \tfrac{1}{2}\upsilon_{Y,i,t}. \tag{27}$$

Equivalently,

$$a_{i,r} \ = \ r_t + \tfrac{1}{2}\varsigma_r^2 \ - \ \tfrac{1}{2}\ \mathrm{Var}_t\big((\lambda_r + \lambda_{i,r})'\Delta g_{t+1} + \Gamma_i'\Delta z_{t+1} + \upsilon_{t+1} + \epsilon_{i,t+1}\big),$$

where we used $\upsilon_{Y,i,t}$ from (26). This is the risk-adjusted drift restriction implied by no-arbitrage under log-normality. Recognizing the difference of the end-to-end levels is obtained by summing the difference equation over time to recover levels, we have, by summing (12) over $s = 0, \dots, t-1$:

$$X_{i,t} - X_{i,0} = \sum_{s=0}^{t-1} \Delta X_{i,s+1} = \sum_{s=0}^{t-1} a_{i,r} + \lambda_{i,r}' \sum_{s=0}^{t-1} \Delta g_{s+1} + \Gamma_i' \sum_{s=0}^{t-1} \Delta z_{s+1} + \sum_{s=0}^{t-1} \epsilon_{i,s+1}.$$

Using telescoping sums and the fact that $\epsilon_{i,s+1} = \Delta u_{i,s+1}$,

$$X_{i,t} = X_{i,0} + a_{i,r}\, t + \lambda_{i,r}'\, g_t + \Gamma_i'\, z_t + u_{i,t} - u_{i,0}.$$

Collect the constants (including $X_{i,0} - u_{i,0}$ and the deterministic trend component implied by $a_{i,r}$) into

$$\mu_{i,r} \ := \ X_{i,0} - u_{i,0} \ + \ \underbrace{\big(\text{risk-adjusted drift term consistent with (27)}\big)}_{\text{absorbed into } \mu_{i,r}}.$$

Thus we can rewrite the level as

$$X_{i,t} \ = \ \mu_{i,r} \ + \ \lambda_{i,r}'\, g_t \ + \ \Gamma_i'\, z_t \ + \ u_{i,t}. \tag{28}$$

By construction, $u_{i,t}$ is stationary ($I(0)$) and we know that, g_t is $I(1)$ and z_t is $I(0)$. Hence (28) is an affine (linear-in-state) representation with $I(1)$ driver g_t, stationary control z_t, and stationary remainder $u_{i,t}$. The constant $\mu_{i,r}$ aggregates initial conditions and the risk-adjusted drift identified by (27). This proves the claim in Lemma 1.

Below, we show a detailed argument that $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

8.5 Computing $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$

Assume the stochastic discount factor M_t follows

$$\frac{dM_t}{M_t} \;=\; -\,r_t\,dt \;-\; \Lambda_t^\top\,dW_t, \tag{29}$$

where r_t is the short rate, W_t is a k-dimensional Brownian motion, and $\Lambda_t \in \mathbb{R}^k$ collects the (vector) prices of risk. By Itô's formula,

$$d\log M_t \;=\; \frac{dM_t}{M_t} \;-\; \frac{1}{2}\left(\frac{dM_t}{M_t}\right)^2.$$

Using (29), note that $\left(\frac{dM_t}{M_t}\right)^2 = (\Lambda_t^\top dW_t)^2 = \|\Lambda_t\|^2\,dt$. Hence

$$d\log M_t \;=\; -\,r_t\,dt \;-\; \Lambda_t^\top dW_t \;-\; \tfrac{1}{2}\,\|\Lambda_t\|^2\,dt. \tag{30}$$

Integrate (30) from t to $t+\Delta$:

$$\log M_{t+\Delta} - \log M_t \;=\; -\int_t^{t+\Delta} r_s\,ds \;-\; \tfrac{1}{2}\int_t^{t+\Delta} \|\Lambda_s\|^2 ds \;-\; \int_t^{t+\Delta} \Lambda_s^\top dW_s. \tag{31}$$

Define the (martingale) shock and its conditional variance:

$$\xi_{t+\Delta} \;\equiv\; \int_t^{t+\Delta} \Lambda_s^\top dW_s, \qquad \varsigma^2 \;\equiv\; \int_t^{t+\Delta} \|\Lambda_s\|^2 ds.$$

Conditionally on $\mathcal{F}_t$, $\xi_{t+\Delta} \sim \mathcal{N}(0,\varsigma^2)$. Then (31) becomes

$$\log M_{t+\Delta} \;=\; \log M_t \;-\; \int_t^{t+\Delta} r_s\,ds \;-\; \tfrac{1}{2}\,\varsigma^2 \;-\; \xi_{t+\Delta}. \tag{32}$$

If we take $\Delta = 1$ and assume $r_s \equiv r_t$ and $\Lambda_s \equiv \Lambda_t$ on $[t, t+1]$, then

$$\int_t^{t+1} r_s\,ds = r_t, \qquad \varsigma^2 = \int_t^{t+1} \|\Lambda_s\|^2 ds = \|\Lambda_t\|^2 \;\equiv\; \varsigma_r^2,$$

and $\xi_{t+1} = \Lambda_t^\top (W_{t+1} - W_t)$. Normalizing $M_t = 1$ (so $\log M_t = 0$), (32) reduces to

$$\log M_{t+1} \; = \; -\, r_t \; - \; \tfrac{1}{2}\,\varsigma_r^2 \; - \; \xi_{t+1}, \tag{33}$$

with $\xi_{t+1} \,|\, \mathcal{F}_t \sim \mathcal{N}(0, \varsigma_r^2)$. Since $\xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2)$, $\mathbb{E}_t[e^{-\xi_{t+1}}] = \exp\left(\tfrac{1}{2}\varsigma_r^2\right)$. Therefore

$$\mathbb{E}_t[M_{t+1}] = \exp\big(- r_t - \tfrac{1}{2}\varsigma_r^2\big)\, \mathbb{E}_t\big[e^{-\xi_{t+1}}\big] = \exp(-r_t),$$

so $-\frac{1}{2}\varsigma_r^2$ is exactly the Jensen correction that pins $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

8.6 Logical Structure of Proposition 2

For clarity, we set out the formal logic underlying Proposition 2 and its testable implication.

(1) *General Statement:* If (A and B) then C:

$$(A \wedge B) \quad \Rightarrow \quad C,$$

where:

A = currencies load similarly on global common factors,

B = crisis volatility rises,

C = long-horizon return correlations converge to one.

(2) *Contrapositive:* By standard logic,

$$\neg C \quad \Rightarrow \quad \neg(A \wedge B).$$

$$\neg(A \wedge B) \quad \Leftrightarrow \quad (\neg A \vee \neg B).$$

(3) *Crisis Simplification:* In systemic crises, volatility almost always rises. Hence $\neg B$ is implausible. This reduces the contrapositive to

$$\neg C \quad \Rightarrow \quad \neg A.$$

(4) *Testable Implication:* If correlations approach one (C), this is consistent with A (currencies load similarly on common factors $\Rightarrow$ reduced autonomy). If correlations remain modest ($\neg C$), then $\neg A$ (factor loadings are not aligned $\Rightarrow$ autonomy persists, cointegration weakens). This provides the empirical testing rule applied in the main text.